



An arc-length control technique for solving quasi-static fracture problems with phase field models and a staggered scheme

J. Zambrano¹ · S. Toro¹ · P. J. Sánchez^{1,2} · F. P. Duda³ · C. G. Méndez¹ · A. E. Huespe¹

Received: 5 May 2023 / Accepted: 15 August 2023 / Published online: 11 September 2023
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2023

Abstract

This paper describes a new arc-length control procedure for tracing the equilibrium curve of brittle fracture problems modeled with a phase field approach. The balance equations of this model are solved with a staggered strategy. The control equation of the arc-length procedure determines the displacement increments during the mechanical stage. The arc-length parameter is interpreted as imposing a given increment of the driving force appearing into the micro-force balance equation. The innovative technique consisting of applying the control equation to the displacement degrees of freedoms (DOFs) of the mechanical stage offers an enhancement over earlier arc-length strategies that focused on controlling the damage DOFs in the micro-force balance equation stage. This advancement enables the phase field approach to handle and simulate a broader range of problems, as demonstrated in this paper. The arc-length parameter is stepwise adjusted to yield a pre-established maximum damage increment in each staggered scheme step. As a consequence, the crack tip advance can be strictly controlled in every step holding bounded the pseudo-time integration error, even using an explicit staggered strategy. This procedure entails moderate computational costs for tracing the complete equilibrium curve, including unstable responses, limit points, snap-backs, etc., with the subsidiary advantage that lack of convergence has never been detected in the tests presented in this paper. Additionally, line search techniques have not been necessary. The proposed arc-length procedure is easily implemented in standard finite element codes, and according to our numerical experiments, it does not significantly increase the computational burden of the original explicit staggered strategy.

Keywords Arc-length control strategy · Phase field models · Staggered methodology · Path-following algorithm · Quasi-static brittle fracture · Unstable crack growth

1 Introduction

In recent years, a vast literature aiming at the numerical simulation of quasi-static evolution of brittle fracture problems through phase field models has been reported. One topic of interest discussed in these works refers to the advantages of using one of the two alternatives, monolithic versus staggered, schemes to solve the coupled equations resulting from the mathematical formulation of the problem.

It is mentioned that the non-convex character of the energy in the space defined by the displacements and phase field variables admits many local minimizers. As pointed out in Bourdin et al. [1] and Farrell and Maurini [2], this feature is the reason why serious convergence issues arise when using monolithic strategies. A notable consequence is that the Newton–Raphson scheme does not necessarily provide descent directions of the related optimization problem. However, according to some authors, a monolithic Newton–Raphson scheme can be used combined with line searches admitting negative step directions, as proposed in Gerasimov and De Lorenzis [3]. Monolithic Quasi-Newton (BFGS) strategies, such as the ones advocated by Kristensen and Martínez-Pañeda [4], have also been reported as a successful technique. It is also significant to mention that numerous variants of monolithic Newton–Raphson schemes have been reported. In this line, we cite the works of Wick [5], Heister

✉ A. E. Huespe
ahuespe@intec.unl.edu.ar

¹ CIMEC-UNL-CONICET, Predio Conicet Dr Alberto Cassano, Santa Fe 3000, Argentina

² GIMNI-UTN-FRSF, Lavaise 610, Santa Fe 3000, Argentina

³ Programa de Engenharia Mecânica - COPPE, Universidade Federal do Rio de Janeiro, Cidade Universitária, Rio de Janeiro, RJ 21941-972, Brazil

et al. [6], Lampron et al. [7], Kopaničáková et al. [8], among others.

Alternatively, a robust staggered solution procedure, reported by Bourdin et al. [1] and followed by Miehe et al. [9], the so-called alternate minimization (AM) method can be used. Nice convergence properties of this methodology have been proven by Bourdin [10], Burke et al. [11] and Brun et al. [12]. In the first stage of an AM methodology, the standard force balance equation¹ is solved with fixed phase field (damage) variables. Subsequently, the micro-force balance equation is solved with fixed displacements. Thus, this staggered scheme relies on solving two sequential convex sub-problems, it is very robust, and the partitioning of the total degrees of freedoms (DOFs) in mechanical and phase field for each sub-problem gives additional computational advantages with respect to the monolithic ones. However, one of the principal deficiencies relies on the fact that its convergence rate is slow.

An additional issue, which is of ubiquitous nature when modeling quasi-brittle fracture problems, is the existence of limit points and snap-backs along the structural equilibrium path. This ingredient requests the use of load control strategies for numerically handling unstable crack growth problems. In the area comprising quasi-brittle fracture problems, an automatic procedure addressed to solve this issue is the control strategy based on limiting the energy dissipation (Gutiérrez [16]), or energy release rate (Verhoosel et al. [17]) in every load step. The adaptation of this control strategy for modeling brittle fracture through the phase field methodology with a staggered scheme has been addressed by Singh et al. [18]. Following the original strategy of Gutierrez et al. and Singh et al. controls in each staggered step the damage variable increment. Therefore, the original AM methodology of Bourdin et al. [19] is modified, the first stage solves a controlled micro-force balance equation with the proposed arc-length technique, and the second stage solves the standard force balance equation.

Also, Bharali et al. [19] employed a monolithic methodology solved through a Newton–Raphson under-relaxed scheme and controlled with an arc-length technique that limits the increment of the damage DOFs. A close monolithic approach to that described by Bharali et al. with an arc-length technique is shown in Börjesson et al. [20]. To the best of our

knowledge, no similar approach has been reported that uses a quasi-Newton scheme controlled by an arc-length technique. Brust and Marcia [21] have presented a procedure for solving large-scale optimization problems with linear equality constraints using quasi-Newton methods. However, the application of this procedure to the area of interest of the present paper is not yet studied.

A procedure for controlling the increments of displacements in the first mechanical sub-problem, i.e., the standard force balance equation, using the AM of Bourdin et al. has been reported previously by the authors, see Duda et al. [15]. In that work, the controlled magnitude is the relative displacement increment between the two nodes opposite the crack tip. Thus, as the crack propagates, the controlling nodes are redefined to follow the crack tip². Also, Wu [23] reported an arc-length strategy addressed to the original AM methodology. Wu proposes a control strategy based on regulating the conventional crack mouth opening displacements (CMOD), crack mouth sliding displacement (CMSD), or a similar magnitude in pre-cracked specimens undergoing fracture. However, in generic crack propagation problems, both control schemes, that of Duda et al. and Wu, are unsuitable. In the first case, it is not possible to know beforehand the crack propagation path, and therefore the nodes to be controlled are unknowns of the problem. In the second case, the CMOD or CMSD magnitudes are not necessarily monotonous increasing variables during the loading process, as shown in the plot of Fig. 3c below. For these specific problems, this feature makes the CMOD an impractical variable to be controlled through an arc-length strategy.

In the present work, we describe an alternative load control procedure for solving the standard force balance equations adapted to the original AM method of Bourdin et al. The control strategy governs the increments of displacements in the first mechanical sub-problem when the damage variables are held fixed. Thus, by construction, this control strategy cannot be based on handling the energy dissipation magnitude of the step because it is null while solving the mechanical sub-problem. Instead, we control the increment of the driving force that appears in the micro-force balance equation. Compared with the arc-length control technique reported by Singh et al. [18], our strategy can be extended to handle loading processes of more general phase field problems including those whose micro-force balance equation and complementarity conditions are solved via optimization techniques with inequality constraints, or problems with several unstable dissipative mechanisms, not only phase field. Section 3.2.2 provides additional details on the novelty and advantages of the present methodology, while Sects. 4.2 and 4.5 describe significant tests that prove these features.

¹ The Euler–Lagrange equations of the variational model derived from the energy minimization of a regularized problem, established in Bourdin et al. [1], references cited therein, and Miehe et al. [9], are the standard force and micro-force balance equations that can be solved with a staggered strategy. However, a general approach to phase field fracture problems that yields similar force and micro-force balance equations without appealing to the existence of a minimum energy law can be followed. This approach has been reported elsewhere. See for, instance, Duda et al. [13], da Silva Jr et al. [14], Duda et al. [15], and references cited therein. Here, this last approach is adopted.

² Close to this approach is the strategy reported by Martínez-Pañeda et al. [22] and references cited therein.

The proposed staggered strategy with the arc-length control procedure is described, implemented, and tested for a specific brittle fracture phase field model, denoted AT2 in Tané et al. [24] combined with the complementarity conditions satisfied either through the energy history field notion, see Miehe et al. [25], and from now on called the compact history function procedure, or through a Bound-Constrained Optimization Problem (BCOP), as reported in Amor et al. [26].

A summary of the paper is the following. Section 2 presents a brief description of the phase field model for brittle fracture. Section 3 describes the control strategy and the staggered scheme for solving the two balance equations of the problem. Also, in this Section, the novelty of this strategy is discussed. In Sect. 4, the proposed control strategy is assessed via numerous tests. Our objective is to show the robustness, accuracy, and ability of the procedure to trace equilibrium curves characterized by abrupt snap-backs. Finally, in Sect. 5, attention is addressed to assess the computational burden demanded by the methodology.

2 Phase field model for brittle fracture

The variational formulation of the phase field model for fracture in solids is briefly described in this Section. We closely follow the mathematical approach discussed by da Silva Jr et al. [14] and Duda et al. [15].

Let us consider a body $\mathcal{B}$ undergoing quasi-static deformations satisfying the small-strain assumption. The boundary of $\mathcal{B}$ is $\partial\mathcal{B}$ with unit normal vector $\mathbf{n}$. Deformation and fracture in $\mathcal{B}$ are described by introducing the displacement vector field, $\mathbf{u}(\mathbf{x}, t)$, and phase field variable, $\varphi(\mathbf{x}, t)$, at time t and $\mathbf{x} \in \mathcal{B}$. The phase field takes values on the interval $[0, 1]$. If $\varphi = 0$ at a point, then it is unfractured. If $\varphi = 1$ at a point, it is fractured. Values of φ between zero and one correspond to partially fractured material. Points satisfying $\varphi = 1$ identify cracks, i.e. a traction-free boundary embedded into the bulk material. Thermodynamic arguments impose the restriction: $\dot{\varphi} \geq 0$. The fields $\mathbf{u}$ and φ are assumed smooth on $\mathcal{B}$.

Both independent primal descriptors, $\mathbf{u}$ and φ , characterize the mechanical state of $\mathcal{B}$ at pseudo-time “ t ”. They are gathered in the vector $\mathbf{U} = \{\mathbf{u}, \varphi\}$. Admissibility conditions for $\mathbf{U}$ are established in the form $\mathbf{U} \in \mathcal{K}$, where the set $\mathcal{K}$ accounts for proper regularity and Dirichlet-type boundary conditions of every descriptor. In particular, we consider displacements, $\mathbf{u}^*$, fixed on a part of the Dirichlet-boundary $\partial\mathcal{B}_D^u$ of $\mathcal{B}$. Virtual variations of $\mathbf{U}$ are denoted $\delta\mathbf{U} = \{\delta\mathbf{u}, \delta\varphi\}$, and their admissibility conditions are defined through $\delta\mathbf{U} \in \mathcal{V}$, where the vectorial space $\mathcal{V}$ accounts for appropriate regularity requirements and the specified restrictions on $\mathbf{U} \in \mathcal{K}$.

The strain tensor, $\boldsymbol{\varepsilon}$, and its virtual variation, $\delta\boldsymbol{\varepsilon}$, are:

$$\boldsymbol{\varepsilon} = \frac{\nabla\mathbf{u} + (\nabla\mathbf{u})^T}{2}, \quad \delta\boldsymbol{\varepsilon} = \frac{\nabla\delta\mathbf{u} + (\nabla\delta\mathbf{u})^T}{2}, \quad (1)$$

respectively, $\nabla(\cdot)$ being the gradient operator.

2.1 Conjugate variables

The external force field per unit volume in $\mathcal{B}$, $\mathbf{b}$, and the external tractions per unit area in $\partial\mathcal{B}_N^u$, $\mathbf{t}^*$, are dual variables to $\mathbf{u}$, where $\partial\mathcal{B}_N^u$ is the Neumann-boundary associated to $\mathbf{u}$. The Cauchy stress tensor $\boldsymbol{\sigma}$, defined in $\mathcal{B}$, and $\boldsymbol{\varepsilon}$ forms a conjugate pair.

The phase field descriptor, φ , has associated three dual variables: the internal micro-force field π per unit volume, the external micro-force γ per unit volume, and the external micro-traction ζ^* per unit area in $\partial\mathcal{B}_N^\varphi$, where $\partial\mathcal{B}_N^\varphi$ is the Neumann-boundary related to φ . However, in concordance with Duda et al., we further assume that the external loading system conjugated to φ is null. Thus, neither γ nor ζ^* is accounted for in the following. Furthermore, a micro-stress vector field $\boldsymbol{\xi}$ is the conjugate variable of the phase field gradient $\nabla\varphi$.

In Sect. 2.3, the constitutive relations for $\boldsymbol{\sigma}$, $\boldsymbol{\xi}$, and for the active counterpart, π_a , of π are introduced.

2.2 Basic laws

We stipulate that deformation and fracture processes in $\mathcal{B}$ must be consistent with the Variational Principle of Virtual Work. The force balance equation in local form and the phase field micro-force balance equation are derived from this Principle, which, together with the free-energy imbalance comprise the kinetic foundation of the theory in da Silva Jr et al. [14] and Duda et al. [15].

2.2.1 Virtual Power Principle

The internal and external virtual powers, $\delta\mathcal{W}^{\text{int}}$ and $\delta\mathcal{W}^{\text{ext}}$, for any part $\mathcal{P}$ of the body $\mathcal{B}$ are defined as follows:

$$\delta\mathcal{W}^{\text{int}} = \int_{\mathcal{P}} (\boldsymbol{\sigma} \cdot \delta\boldsymbol{\varepsilon} + \boldsymbol{\xi} \cdot \nabla\delta\varphi + \pi \delta\varphi) d\mathcal{P}, \quad (2)$$

$$\delta\mathcal{W}^{\text{ext}} = \int_{\mathcal{P}} (\mathbf{b} \cdot \delta\mathbf{u}) d\mathcal{P} + \int_{\partial\mathcal{P}} (\mathbf{t}^* \cdot \delta\mathbf{u}) d(\partial\mathcal{P}), \quad (3)$$

where $\mathbf{t}^*$ is the traction vector on the boundary $\partial\mathcal{P}$ of $\mathcal{P}$.

The Virtual Power Principle stipulates that: $\delta\mathcal{W}^{\text{int}} = \delta\mathcal{W}^{\text{ext}} \quad \forall \delta\mathbf{U} \in \mathcal{V}$. Admitting independent variations of displacements, we derive integral balance equations, which can be localized using a standard procedure. The resulting local

form of the force balance equation for the body $\mathcal{B}$ is given by:

$$\begin{aligned} \text{Div} \boldsymbol{\sigma} + \mathbf{b} &= \mathbf{0}, & \forall \mathbf{x} \in \mathcal{B}, \\ \boldsymbol{\sigma} \cdot \mathbf{n} &= \mathbf{t}^*, & \forall \mathbf{x} \in \partial \mathcal{B}_N^u. \end{aligned} \quad (4)$$

with $\mathbf{n}$ being the unit normal vector to $\partial \mathcal{B}$. Also, assuming variations for the phase field primal descriptors, and after applying a standard localization procedure, we obtain the micro-force balance equation:

$$\begin{aligned} \text{Div} \boldsymbol{\xi} - (\pi_a + \pi_r) &= 0, & \forall \mathbf{x} \in \mathcal{B}, \\ \boldsymbol{\xi} \cdot \mathbf{n} &= \mathbf{0}, & \forall \mathbf{x} \in \partial \mathcal{B}_N^\varphi, \end{aligned} \quad (5)$$

where we have replaced the internal micro-force π by active and reactive components, π_a and π_r , respectively. The reactive component, in combination with the complementarity conditions: $\pi_r \dot{\varphi} = 0$, $\pi_r \leq 0$, and $\dot{\varphi} \geq 0$, is introduced in the model to guarantee the energy imbalance thermodynamic restriction: $\dot{\varphi} \geq 0$, see details in Duda et al. [15].

2.3 Constitutive theory

The free energy, ψ , is assumed to be defined in terms of the set of variables $\{\boldsymbol{\varepsilon}, \varphi, \nabla \varphi\}$:

$$\psi = \hat{\psi}(\boldsymbol{\varepsilon}, \varphi, \nabla \varphi), \quad (6)$$

and is partitioned into two parts:

$$\psi(\boldsymbol{\varepsilon}, \varphi, \nabla \varphi) = \hat{\psi}_e(\boldsymbol{\varepsilon}, \varphi) + \hat{\psi}_f(\varphi, \nabla \varphi), \quad (7)$$

with

$$\begin{aligned} \hat{\psi}_e(\boldsymbol{\varepsilon}, \varphi) &= \left((1 - \varphi)^2 \right) \hat{\psi}_e^{pos}(\boldsymbol{\varepsilon}) + \hat{\psi}_e^{neg}(\boldsymbol{\varepsilon}), \\ \hat{\psi}_e^{pos}(\boldsymbol{\varepsilon}) &= \frac{1}{2} \left(\lambda \langle \text{tr} \boldsymbol{\varepsilon} \rangle^2 + 2\mu \boldsymbol{\varepsilon}^{pos} : \boldsymbol{\varepsilon}^{pos} \right), \\ \hat{\psi}_e^{neg}(\boldsymbol{\varepsilon}) &= \frac{1}{2} \left(\lambda \langle -\text{tr} \boldsymbol{\varepsilon} \rangle^2 + 2\mu \boldsymbol{\varepsilon}^{neg} : \boldsymbol{\varepsilon}^{neg} \right), \\ \boldsymbol{\varepsilon}^{pos} &= \sum_{k=1}^3 \langle \varepsilon_k \rangle \mathbf{e}_k \otimes \mathbf{e}_k, \quad \boldsymbol{\varepsilon}^{neg} = \sum_{k=1}^3 -\langle -\varepsilon_k \rangle \mathbf{e}_k \otimes \mathbf{e}_k, \end{aligned} \quad (8)$$

where λ and μ are the Lamé parameters, ε_k and $\mathbf{e}_k$ are the k -th eigenvalue and eigenvector of $\boldsymbol{\varepsilon}$, respectively. The angle bracket operator, $\langle \cdot \rangle$, is the positive part of the argument. The partition of $\hat{\psi}_e$ defined in Eq. (8) is taken from Miehe et al. [9].

The second term in Eq. (7) results:

$$\hat{\psi}_f(\varphi, \nabla \varphi) = G_c \left(\frac{\varphi^2}{2\ell} + \frac{\ell}{2} |\nabla \varphi|^2 \right), \quad (9)$$

with G_c and ℓ denoting the fracture energy and the phase field order parameter, respectively.

Thus, the conjugate variables $\boldsymbol{\sigma}$, $\boldsymbol{\xi}$, π_a , are determined through the partial derivative of the free energy densities with respect to their corresponding dual variables:

$$\begin{aligned} \boldsymbol{\sigma} &= \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}} = \left((1 - \varphi)^2 \right) (\lambda \langle \text{tr} \boldsymbol{\varepsilon} \rangle \mathbf{I} + 2\mu \boldsymbol{\varepsilon}^{pos}) \\ &\quad + (-\lambda \langle -\text{tr} \boldsymbol{\varepsilon} \rangle \mathbf{I} + 2\mu \boldsymbol{\varepsilon}^{neg}), \end{aligned} \quad (10)$$

$$\boldsymbol{\xi} = \frac{\partial \psi}{\partial \nabla \varphi} = G_c \ell \nabla \varphi, \quad (11)$$

$$\pi_a = \frac{\partial \psi}{\partial \varphi} = \frac{G_c}{\ell} \varphi - 2(1 - \varphi) \hat{\psi}_e^{pos}(\boldsymbol{\varepsilon}), \quad (12)$$

respectively. In Eq. (10), $\mathbf{I}$ is the second order identity tensor.

In view of (11) and (12), we rewrite (5) as:

$$\begin{aligned} G_c \ell \nabla^2 \varphi - \frac{G_c}{\ell} \varphi + 2(1 - \varphi) \hat{\psi}_e^{pos}(\boldsymbol{\varepsilon}) - \pi_r &= 0, \\ \text{subjected to: } \pi_r \leq 0, \quad \dot{\varphi} \geq 0, \quad \dot{\varphi} \pi_r &= 0 \end{aligned} \quad (13)$$

being $\nabla^2(\cdot)$ the Laplacian operator. Considering the staggered scheme that we propose in sub-Sect. 3.1, this complementarity problem can be solved using the following strategies:

(i) Reformulating the Eq. (13) by introducing the compact history function, $\mathcal{H}(t)$, such as proposed by Miehe et al. [25]. According with such strategy, π_r is defined as:

$$\pi_r = -2(1 - \varphi) (\mathcal{H}(t) - \hat{\psi}_e^{pos}(\boldsymbol{\varepsilon}, t)), \quad (14)$$

where $\mathcal{H}(t)$ results:

$$\mathcal{H}(t) = \max_s h(s), \quad s \in [0, t], \quad (15)$$

with:

$$h(t) := \hat{\psi}_e^{pos}(\boldsymbol{\varepsilon}, t). \quad (16)$$

By construction, the last term within the parenthesis in (14) is greater than or equal to zero. Therefore, the inequality $\pi_r \leq 0$ is automatically satisfied.

(ii) Solving the following inequality-constrained optimization problem with $\boldsymbol{\varepsilon}$ fixed, as proposed by Amor et al. [26]:

$$\min_{1 \geq \varphi \geq 0; \dot{\varphi} \geq 0} \int_{\mathcal{B}} \psi(\boldsymbol{\varepsilon}, \varphi, \nabla(\varphi)) d\mathcal{B}. \quad (17)$$

The functional ψ is given by expressions (7)–(9). Considering the increments of pseudo-time Δt , the pseudo-times $(t - \Delta t)$ and t and defining $\varphi(t) = \varphi(t - \Delta t) + \Delta\varphi$, we rewrite this problem as follows

$$\min_{\Delta\varphi \geq 0; \, 1 \geq \varphi(t)} \int_{\mathcal{B}} \hat{\psi}(\Delta\varphi, \nabla(\Delta\varphi)) d\mathcal{B}, \quad (18)$$

where $\hat{\psi} = \psi(\boldsymbol{\varepsilon}(t), \varphi(t), \nabla(\varphi(t)))$.

(iii) Imposing the inequality constraints via penalization or Lagrangian Multiplier techniques, as studied by Gerasiimov and De Lorenzis [27] and references cited therein.

3 Discrete equation system of the phase field model

After reformulating the force and micro-force balance Eqs. (4–5) into variational weak forms, replacing the constitutive expressions (10–12) into these weak forms and finally implementing a finite element approach for both balance equations, a rather standard procedure whose details have been described in Duda et al. [15], we finally express the resulting discrete equation system at pseudo-time t as follows:

(1) Discrete force balance equations:

$$\mathbf{g}(\mathbf{u}(t), \boldsymbol{\varphi}(t), \mathbf{f}^{ext}(t)) = \mathbf{0}; \quad (19)$$

(2) Discrete micro-force balance equations:

$$\mathbf{J}(\mathbf{u}(t), \boldsymbol{\varphi}(t)) = \mathbf{0}, \quad (\dot{\varphi}_j \geq 0, \, \forall j); \quad (20)$$

where the variables $\mathbf{u}(t), \boldsymbol{\varphi}(t)$ are now reinterpreted as the vectors gathering the displacement and phase field nodal values in the finite element mesh. In the equation (19), $\mathbf{f}^{ext}(\mathbf{u}^*, \mathbf{t}^*)$ denotes the vector of external actions that includes the tractions (Neumann boundary conditions) as well as the non-homogeneous displacements (Dirichlet boundary conditions) imposed on the boundary. Complementarity conditions for every component j -th of the vector $\boldsymbol{\varphi}$ are implicit in (20).

Next, we assume that the loading condition can be represented through a time-dependent scalar parameter, $\lambda(t)$, that determines the magnitude of the external action imposed on the body at time t , $\mathbf{f}^{ext}(\mathbf{u}^*, \mathbf{t}^*) = \lambda \hat{\mathbf{f}}^{ext}$, where $\hat{\mathbf{f}}^{ext}$ denotes a unit vector. Thus, $\lambda(t)$ identifies the external load magnitude at pseudo-time t .

3.1 Staggered Scheme

The staggered scheme to integrate the phase field problem equations at step $(n + 1)$ uses a sequential solution of both balance equations as follows:

- First, given λ_{n+1} at step $n + 1$, solve the standard discrete force balance equation with fixed damage variables whose values are those at the step n :

$$\mathbf{g}(\mathbf{u}_{n+1}, \boldsymbol{\varphi}_n, \lambda_{n+1}) = \mathbf{0}. \quad (21)$$

- Second, the phase field micro-force balance Eq. (20) is solved with fixed external loads, and the displacements are those obtained in the previous stage:

$$\mathbf{J}(\mathbf{u}_{n+1}, \boldsymbol{\varphi}_{n+1}) = \mathbf{0}, \quad (1 \geq (\varphi_j)_{n+1} \geq (\varphi_j)_n). \quad (22)$$

This process can be iterated by solving (21) with the value of the variable $\boldsymbol{\varphi}$ obtained as solution of (22). Following the denomination given by Lu et al. [28], we call it an explicit staggered scheme when no iteration is prescribed. This procedure is an alternative to an implicit staggered strategy where iterations of both stages, i.e., the sequential solutions of (21) and (22), is repeated in a given loading step until reaching the error convergence in a given norm. Explicit and implicit procedures could yield an uncontrolled advance of the crack in a one load step.

In the following, we show that an explicit staggered scheme with controlled error via an arc-length procedure could be a convenient technique for solving phase field brittle fracture problems.

3.2 Arc-length procedure with step length adaptation

A procedure for controlling the load increment along the equilibrium path in phase field models can be derived from energy-based control techniques reported in the literature, see [16–18, 29].

The strategy consists of introducing a monotonously increasing arc-length parameter, s , governing the evolution of the load factor and displacements in the problem (21) jointly with an additional scalar control equation defining the magnitude of the increment of s . We recall that problem (21) is characterized by holding fixed the damage variables, and therefore, the dissipation is null in this stage.

The reformulated problem consists of solving:

$$\mathbf{g}(\mathbf{u}_{(s_{n+1})}, \boldsymbol{\varphi}_n, \lambda_{(s_{n+1})}) = \mathbf{0}, \quad (23)$$

$$r(\mathbf{u}_{(s_{n+1})}, \lambda_{(s_{n+1})}, s_{n+1}) = 0, \quad (24)$$

where $\mathbf{u}_{n+1}$ and λ_{n+1} are now reinterpreted as implicit functions of s_{n+1} related through the control equation (24) and defined in the following.

Let us consider the term:

$$\mathcal{F}^{\text{driv}}(\mathbf{u}(t), \varphi_n) = 2(1 - \varphi_n)\mathcal{H}(\mathbf{u}(t)), \quad (25)$$

such that: $\dot{\mathcal{F}}^{\text{driv}} = 2(1 - \varphi_n)\dot{\mathcal{H}}$ and $\mathcal{H}(\mathbf{u}, t)$ satisfies the conditions (15)–(16) at pseudotimes t and $t + dt$. Then, rewriting Eqs. (13) and (14) in rates³ and operating, they result in:

$$-G_c \ell \nabla^2 \dot{\varphi} + \left(\frac{G_c}{\ell} + 2\mathcal{H} \right) \dot{\varphi} = \underbrace{2(1 - \varphi)\dot{\mathcal{H}}}_{\dot{\mathcal{F}}^{\text{driv}}}, \quad (26)$$

where, we interpret $\dot{\mathcal{F}}^{\text{driv}}$ as being the rate of a driving force governing the evolution of φ . Inspired by the role played by this term, we propose a driving force control-based strategy, reminiscent from May et al. [29], in where (24) is defined as follows:

$$\begin{aligned} r(\mathbf{u}(s_{n+1}), \lambda(s_{n+1}), s_{n+1}) \\ = \int_{\mathcal{B}} \mathcal{F}^{\text{driv}}(\mathcal{H}_{n+1}, \varphi_n) d\mathcal{B} - s_{n+1} = \\ = \int_{\mathcal{B}} 2(1 - \varphi_n)\mathcal{H}_{n+1} d\mathcal{B} - s_{n+1} = 0. \end{aligned} \quad (27)$$

Based on these expressions, we use a Newton–Raphson strategy for solving (23) and (27) at step $n + 1$ as follows:

$$\mathbf{K} \Delta \mathbf{u} + \partial_{\lambda} \mathbf{g} \Delta \lambda = -res_u, \quad (28)$$

$$\int_{\mathcal{B}} 2(1 - \varphi_n) \Delta \mathcal{H} d\mathcal{B} - \Delta s = -res_s, \quad (29)$$

where $\mathbf{K}$ is the stiffness matrix, i.e. $\mathbf{K} = \partial_{\mathbf{u}} \mathbf{g}$, the residue of Eqs. (23) and (27) are denoted res_u and res_s , respectively, and $\Delta(\cdot) = (\cdot)_{n+1} - (\cdot)_n$. At every load step, the increment Δs is adjusted according to $\Delta \varphi$, as explained in next sub-Section.

Recalling expressions (15) and (16), the term $\mathcal{H}_{n+1}$ at step $n + 1$ is identified as:

$$\mathcal{H}_{n+1} = \begin{cases} \hat{\psi}_e^{pos}(\mathbf{e}_{n+1}) = \frac{1}{2}(\mathbf{e}_{n+1} \cdot \boldsymbol{\sigma}_{n+1}^{pos}) & \text{if } \hat{\psi}_e^{pos}(\mathbf{e}_{n+1}) > \hat{\psi}_e^{pos}(\mathbf{e}_n) \\ \mathcal{H}_n & \text{if } \hat{\psi}_e^{pos}(\mathbf{e}_{n+1}) \leq \hat{\psi}_e^{pos}(\mathbf{e}_n) \end{cases}, \quad (30)$$

where we define

$$\boldsymbol{\sigma}_{n+1}^{pos} = \begin{cases} (\lambda \langle \text{tr } \mathbf{e}_{n+1} \rangle \mathbf{I} + 2\mu \mathbf{e}_{n+1}^{pos}) & \text{if } \hat{\psi}_e^{pos}(\mathbf{e}_{n+1}) > \hat{\psi}_e^{pos}(\mathbf{e}_n) \\ 0 & \text{if } \hat{\psi}_e^{pos}(\mathbf{e}_{n+1}) \leq \hat{\psi}_e^{pos}(\mathbf{e}_n) \end{cases},$$

³ The finite element approach of this equation is described in Duda et al. [15].

(31)

as well as the first order approximation of $\dot{\mathcal{H}}$: $\Delta \mathcal{H} = \mathcal{H}(\mathbf{e}_{n+1}) - \mathcal{H}(\mathbf{e}_n)$, it results:

$$\Delta \mathcal{H} = \begin{cases} \frac{1}{2}(\Delta \mathbf{e} \cdot \boldsymbol{\sigma}_n^{pos} + \mathbf{e}_n \cdot \Delta \boldsymbol{\sigma}^{pos}) & \text{if } \hat{\psi}_e^{pos}(\mathbf{e}_{n+1}) > \hat{\psi}_e^{pos}(\mathbf{e}_n) \\ 0 & \text{if } \hat{\psi}_e^{pos}(\mathbf{e}_{n+1}) \leq \hat{\psi}_e^{pos}(\mathbf{e}_n) \end{cases}. \quad (32)$$

Also, from the identity $\dot{\mathcal{H}} = \dot{\mathbf{e}} \cdot \boldsymbol{\sigma}^{pos}$, we consider the following approximate expression: $\Delta \mathcal{H} \approx \Delta \mathbf{e} \cdot \boldsymbol{\sigma}_{n+1}^{pos}$. Under such conditions, the first term on the left hand side of Eq. (29) is approximated by:

$$\begin{aligned} \int_{\mathcal{B}} 2(1 - \varphi_n) \Delta \mathcal{H} d\mathcal{B} \\ \approx \int_{\mathcal{B}} 2(1 - \varphi_n) \Delta \mathbf{e} \cdot \boldsymbol{\sigma}_{n+1}^{pos} d\mathcal{B} = \Delta \mathbf{u} \cdot \mathbf{F}_{int}^{pos}, \end{aligned} \quad (33)$$

where we have defined the positive internal force vector:

$$\mathbf{F}_{int}^{pos} = \int_{\mathcal{B}} 2(1 - \varphi_n) \mathbf{B}^T \boldsymbol{\sigma}_{n+1}^{pos} d\mathcal{B}. \quad (34)$$

with $\mathbf{B}$ being the conventional strain–displacement matrix.

Finally, taking into account (33), Eqs. (28) and (29) can be solved to obtain:

$$\Delta \mathbf{u} = \Delta \mathbf{u}_I + \Delta \lambda \Delta \mathbf{u}_{II}, \quad (35)$$

where

$$\Delta \mathbf{u}_I = -\mathbf{K}^{-1} res_u, \quad \Delta \mathbf{u}_{II} = -\mathbf{K}^{-1} \partial_{\lambda} \mathbf{g}, \quad (36)$$

$$\Delta \lambda = \frac{(\Delta s - res_s) - \Delta \mathbf{u}_I \cdot \mathbf{F}_{int}^{pos}}{\Delta \mathbf{u}_{II} \cdot \mathbf{F}_{int}^{pos}}. \quad (37)$$

The vector $\mathbf{F}_{int}^{pos}$ is evaluated at each Newton–Raphson iteration of the step $n + 1$.

When expression (33) is equal to zero, the arc-length strategy switches to an alternative control procedure, as discussed in May et al. [29]. However, this situation has never occurred in the tests assessed in Sect. 4.

3.2.1 Adjust of the arc-length parameter in terms of $\Delta \varphi$

The arc-length parameter, Δs , at step $n + 1$ is adjusted in terms of the phase field variable increment at step n as follows:

$$\Delta s_{n+1} = \Delta s_n \left(\frac{\Delta \hat{\varphi}^{opt}}{\|(\boldsymbol{\varphi}_n - \boldsymbol{\varphi}_{n-1})\|_{\infty}} \right)^{1/2} \leq \Delta s^{max}, \quad (38)$$

where $\Delta \hat{\varphi}^{opt}$ and Δs^{max} are two parameters of the algorithm representing the maximum desired phase field increment per staggered scheme step and the maximum allowed size of

the arc-length, respectively. This expression is adopted by following the same idea reported by Crisfield [30], formula (9.41), pp. 287, to adjust the arc-length control parameter but written in terms of $\Delta\hat{\varphi}^{opt}$ and $\Delta\varphi = (\|\varphi_n - \varphi_{n-1}\|_\infty)$ instead of using, as proposed by Crisfield, the desired number of iterations required by the iterative solver and the number of iterations demanded by the last converged step, respectively.

The proposed methodology follows the strategy of adjusting Δs with expression (38), combined with an explicit staggered scheme, i.e., we evaluate only one mechanical and one damage stage per step in the staggered scheme. Furthermore, we calculate the mechanical stage solutions with a tight error tolerance (less than 1×10^{-7}).

The algorithmic performance attained with this strategy is specifically assessed in Sect. 5.

3.2.2 Novelty of the proposed arc-length control strategy

Previous works, such as Bharali et al. [19], Börjesson et al. [20], and Singh et al. [18] have utilized an arc-length procedure with a control equation that limits the increments of the phase field variable. In particular, Singh et al. describe an arc-length procedure combined with a staggered scheme. Our strategy offers an alternative approach whose advantages are discussed in the following.

Two arc-length strategies combined with staggered schemes

Let us reconsider the staggered scheme described with the Eqs. (21) and (22) and the control equation indicated through: $r = 0$, where r is the control function whose argument depends on different variables, according to each arc-length strategies.

- In the first strategy, such as reported by Singh et al. [18], a loading control procedure based on prescribing the increment of the fracture surface area through equation: $r_{dam} = 0$ is proposed. Then, the micro-force balance equation is solved in combination with the control equation:

$$\begin{aligned} J(\mathbf{u}_n, \varphi(s_{n+1}), \lambda(s_{n+1})) &= \mathbf{0} ; \\ (1 \geq \varphi_j(s_{n+1}) \geq \varphi_j(s_n), \forall j) , \end{aligned} \quad (39)$$

$$r_{dam}(\varphi(s_{n+1}), \lambda(s_{n+1}), s_{n+1}) = 0 , \quad (40)$$

with $\mathbf{u}_n$ hold fixed. The staggered scheme follows by solving the discrete force balance equations to find $\mathbf{u}_{n+1}$. Thus, according to this strategy, there is not an explicit relationship between $\mathbf{u}_{n+1}$ and s_{n+1} in each step.

- The present strategy, as shown above, proposes a control load procedure prescribing the increment of the history variable through equation $r_{mech} = 0$. Thus, the equations solved in the mechanical stage are:

$$\mathbf{g}(\mathbf{u}(s_{n+1}), \varphi_n, \lambda(s_{n+1})) = \mathbf{0} , \quad (41)$$

$$r_{mech}(\mathbf{u}(s_{n+1}), \lambda(s_{n+1}), s_{n+1}) = 0 , \quad (42)$$

with φ_n hold fixed. The staggered scheme follows by solving the discrete micro-force balance equations to find φ_{n+1} . By construction, notice that dissipation due to phase field evolution is null in equations (41) and (42).

The two control strategies, applied at different stages of the staggered scheme, have significant algorithmic implications when solving certain types of phase field problems, for example:

- Phase field methodologies that solve the micro-force balance equation using a bound-constrained optimization problem, as formulated in (18), can be easily integrated into our arc-length strategy using the control Eq. (27). A numerical test that demonstrates the feasibility and effectiveness of the procedure in this situation is presented in Sect. 4.2. Other methodologies that impose inequality constraints on the micro-force balance equation via penalization or Lagrangian Multiplier techniques can also be similarly integrated into our approach. In all these methodologies, the common feature leveraged by our procedure is that the increment of the driving force term, $\dot{\mathcal{F}}^{driv}$, governing the damage increment, though different in every methodology, can be closely controlled with the same Eq. (27). On the other hand, when the arc-length strategy limits the phase field increments in the micro-force balance equation, the problem (18) have to be solved monolithically in the extended space of variables (φ, λ) combined with the control equation $r_{dam} = 0$. In this case, the problem to be solved can no longer be viewed as the optimality condition of a minimization problem in the increased space of variables. Thus, conventional constrained optimization algorithms are inapplicable.
- Secondly, in problems where the damage mechanism, modeled through the phase field variable, coexists with other dissipative mechanisms that induce instability and are related to $\mathbf{u}$, such as happens in plasticity, cohesive interfaces, etc., the arc-length strategy has to manage the loading process whichever the mechanism inducing instability is.

As discussed earlier, our strategy allows for controlling the competition between several unstable mechanisms by extending the definition of the control function r_{mech} , as demonstrated in the numerical assessment of Sect. 4.5. However, with the alternative arc-length strategy, it is challenging to include the control of displacement DOFs that experience instability by extending the definition of the control function r_{dam} .

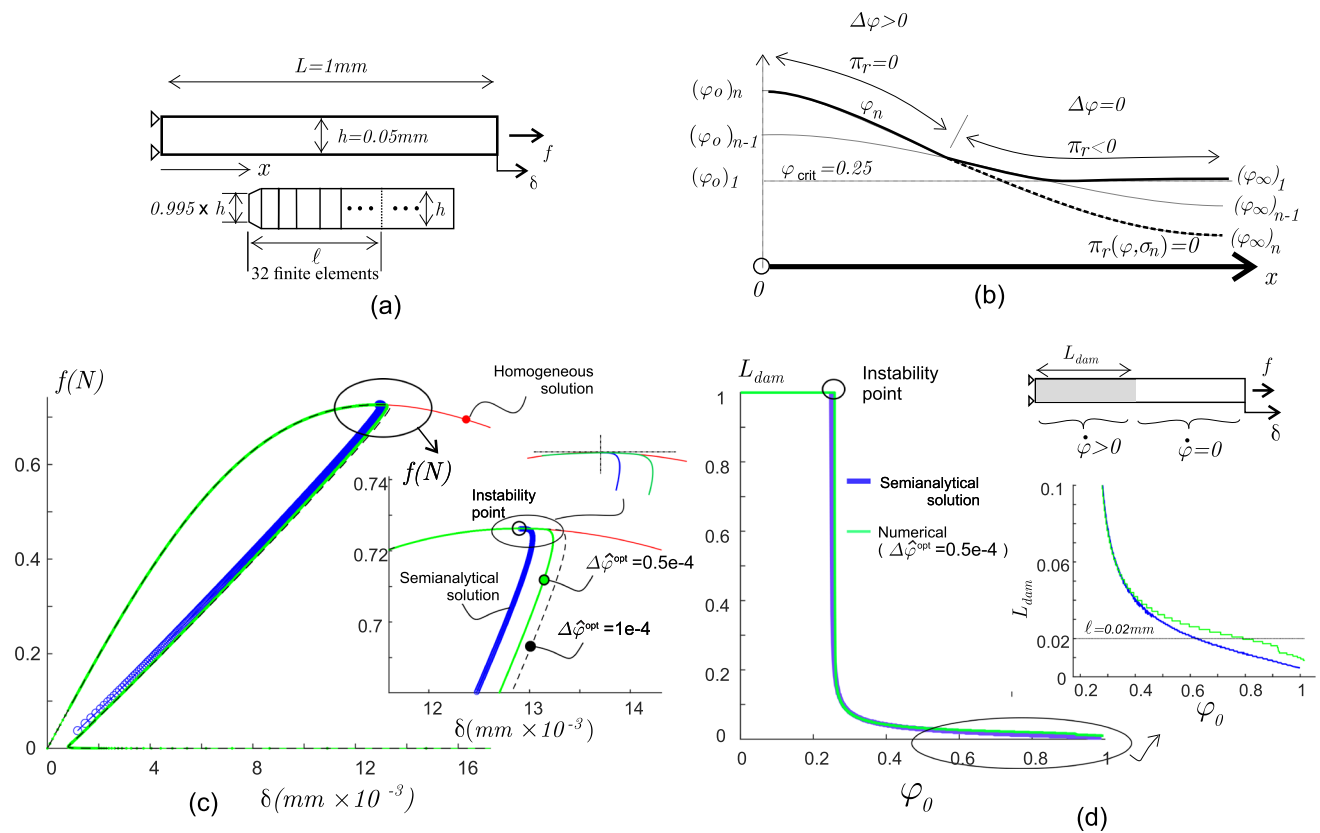


Fig. 1 Bar under tensile load. **a** Geometrical perturbation and Finite Element model detail. **b** Semi-analytical procedure satisfying the complementarity conditions. **c** Force versus displacement plot. Curves corresponding to the homogeneous and localized solutions have hor-

izontal tangent lines at the limit point. See further details in the inset. **d** Size of the localization zone, L_{dam} , in terms of the maximum damage, φ_0 , in successive load steps

4 Numerical assessments

With the numerical assessments in this Section, we achieve two objectives. Firstly, we evaluate the performance of the previously outlined arc-length control strategy. The seven tests display unstable crack propagation under displacement or force control demonstrating the necessity of using an arc-length procedure for controlling the crack tip advance and the numerical integration error. Secondly, we show the range of problems that can be assessed with the present arc-length technique combined with a staggered strategy. Thus, the Double-Edged Notched Test (DENT), in Sect. 4.2, is solved by appealing to the compact history function technique as well as to the Bound-Constrained Optimization Problem (BCOP). Furthermore, the last test in sub-Sect. 4.5 displays multiple instability mechanisms, not only those caused by the phase field evolution. This test highlights the capacity of the arc-length procedure for controlling multiple instabilities, including those produced by the phase field evolution.

The computational performance of the algorithm implemented to solve these tests are further analysed in the following Sect. 5.

As general comments describing the following seven tests, we mention that the pre-existent cracks in the DENT specimen and the Notched Plate with Hole Test (NPHT) of subsection 4.3 are modeled with a finite element mesh of unconnected elements. We also consider two different initial conditions for the phase field variable. In general, we assume that $\varphi(\mathbf{x}) = 0 \forall \mathbf{x} \in \mathcal{B}$. However, we also report alternative solutions with a no-null initial condition of φ , the initial nodal values of φ are constrained to 1 in the nodes on the notch root, a procedure used in Singh et al. [18] and here denoted as Phase Field Tip-Enrichment (PFTE). Plane strain conditions are also assumed in all tests, and the bar problem in sub-Sect. 4.1 is simulated with a Poisson's ratio $\nu = 0$.

4.1 Instability analysis of a uniform tensile bar

We analyze the unstable response of a uniform bar under tensile load. A semi-analytical solution of the problem, sim-

ilar to the one reported by Pham and Marigo [31], is briefly described. Afterward, we compare the numerical solutions obtained with the present control strategy using a small perturbation of the bar section triggering the non-homogeneous solution. The arc-length control strategy plays a role in transitioning smoothly through the instability region.

4.1.1 Semi-analytical solution of an infinite bar under tensile stress

Let us consider a 1D bar of infinite length placed along the x -axis undergoing a tensile load. The tensile stress is σ . The balance Eq. (4) particularized for this problem, neglecting body forces, results in $\sigma_{,x} = 0$. The sub-index identifies the derivative with respect to the variable x . Similarly, the first three terms on the left-hand side of the micro-force balance Eq. (13) results:

$$\pi_r(\varphi, \sigma) = G_c \ell \varphi_{,xx} - \frac{G_c}{\ell} \varphi + \frac{\sigma^2}{E(1-\varphi)^3} \leq 0, \quad \dot{\varphi} \geq 0, \quad \dot{\varphi} \pi_r = 0, \quad (43)$$

where we have used the identity: $\hat{\psi}_e^{pos}(\boldsymbol{\varepsilon}) = \sigma^2/(2E(1-\varphi)^4)$, with E the Young modulus of the bar.

One homogeneous solution, φ^h , satisfying $\varphi_{,x}^h = 0$ is determined from the condition $\pi_r = 0$ and results in:

$$\sigma^h = \sqrt{\frac{G_c E}{\ell} \varphi^h (1 - \varphi^h)^3}. \quad (44)$$

The strain is determined from the constitutive Eq. (10) particularized for the 1D problem:

$$\varepsilon^h = \sqrt{\frac{G_c}{E \ell} \frac{\varphi^h}{(1 - \varphi^h)}}. \quad (45)$$

The solution $\sigma^h(\varepsilon^h)$ with $\varphi^h \in [0, 1]$ becomes unstable at the point of maximum stress $\sigma^{h,crit}$ obtained for $\varphi^{crit} = 0.25$, independently of the material properties. We also consider the non-homogeneous solution of equation $\pi_r(\varphi, \sigma) = 0$, which is reformulated as follows:

$$\left(-(\varphi_{,x})^2 + \frac{\varphi^2}{\ell^2} - \frac{\sigma^2}{G_c E \ell (1 - \varphi)^2} \right)_{,x} = 0, \quad (46)$$

whose solution is:

$$-(\varphi_{,x})^2 + \frac{\varphi^2}{\ell^2} - \frac{\sigma^2}{G_c E \ell (1 - \varphi)^2} - k_\sigma = 0, \quad (47)$$

being k_σ a constant term determined from the boundary condition at $x \rightarrow \infty$, where $\varphi_{,x} \rightarrow 0$, $\sigma = \sigma_\infty$ and $\varphi(x \rightarrow \infty) = \varphi_\infty$. Thus, k_σ results in:

$$k_\sigma = \frac{\varphi_\infty^2}{\ell^2} - \frac{\sigma_\infty^2}{G_c E \ell (1 - \varphi_\infty)^2}. \quad (48)$$

With these values of k_σ and σ_∞ , there is a phase field solution, $\varphi(x)$, with $\varphi(0) = \varphi_0 \geq \varphi_\infty$ and $\varphi_{,x}(0) = 0$, satisfying (47). Thus, Eq. (47) can be integrated as follows:

$$x(\varphi^*) = \int_{\varphi_0}^{\varphi^*} \left(\frac{\varphi^2}{\ell^2} - \frac{\sigma_\infty^2}{G_c E \ell (1 - \varphi)^2} - k_\sigma \right)^{-1/2} d\varphi, \quad (49)$$

where $\varphi^* \geq \varphi_\infty$, and $x(\varphi^*)$ is the coordinate of the bar in where $\varphi(x) = \varphi^*$.

The semi-analytical procedure for determining the non-homogeneous solution along the loading process consists of parameterizing the non-homogeneous equilibrium path with the stress σ . In successive discrete steps (N_{steps} is the number of steps), we define $\sigma_n \in [0, \sigma^{crit}]$, starting at step 1 with $\sigma_1 = \sigma^{crit}$ and decreasing the stress until reaching $\sigma_{N_{steps}} = 0$. Thus, given σ_n , we determine the corresponding terms $(\varphi_0)_n$, $(\varphi_\infty)_n$ for $(\sigma_\infty)_n = \sigma_n$, and numerically integrate the curve $\pi_r(\varphi(x), \sigma_n) = 0$ using expression (49). If the solution $\varphi(x) > \varphi_{n-1}(x)$, we take $\varphi_n(x) = \varphi(x)$, if not, we take $\varphi_n(x) = \varphi_{n-1}(x)$. This semi-analytical procedure is sketched in Fig. 1-b.

4.1.2 Finite element solution and the arc-length control strategy

The finite element solution of the 1D test is assessed with a unit length bar. The model parameters are Young modulus $E = 100$ MPa, fracture energy $G_c = 0.001$ N/mm, and characteristic length: $\ell = 0.02$ mm. A perturbation of the bar area is established, as depicted in Fig. 1a. The region of the bar with coordinates $0 \leq x \leq \ell$ is modeled with 32 finite elements with sizes becoming smaller as $x \rightarrow 0$. A fine mesh in this region is necessary to assess the tendency of the localization zone thickness as the damage increases to 1.

Figure 1c. depicts plots of force f versus displacement δ . We compare the semi-analytical solution with two finite element solutions obtained with the arc-length procedure. The parameter $\Delta\hat{\varphi}^{opt}$ of the arc-length procedure are 10^{-4} , and 0.5×10^{-4} , respectively, enforcing very short step increments.

We identify the bar localization zone as the points satisfying $\dot{\varphi} > 0$. The localization zone size, L_{dam} , changes during the loading process. Figure 1d shows L_{dam} versus the maximum damage, $(\varphi_0)_n$, as the loading process progresses. This Figure plots the solutions of the finite element and semi-analytical models. Details are shown in the inset. Observe that

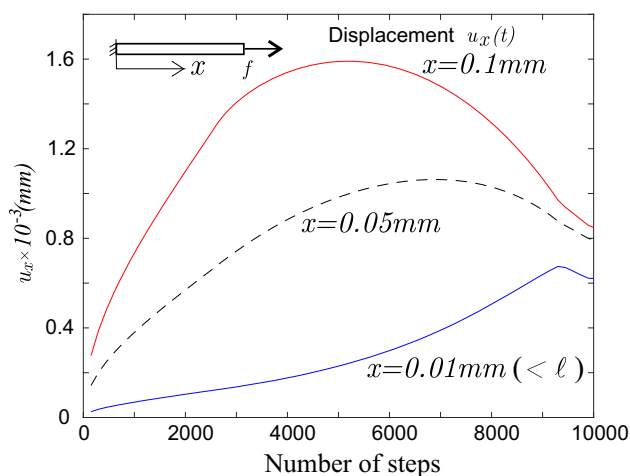


Fig. 2 Bar under tensile load. Displacements, u_x , of bar points at coordinates: $x = 0.1$ mm, $x = 0.05$ mm, and $x = 0.01$ mm versus number of load step. Solution with $\Delta\hat{\phi}^{opt} = 0.5 \times 10^{-4}$

the localization zone width tends to zero as the damage grows to 1. This result means that all nodal displacements, except the ones closer to $x = 0$, display a backtrack behaviour, as shown in Fig. 2. This feature makes this problem particularly difficult to attain solutions with loading techniques based on displacement control.

4.2 DENT panel test under uniform displacement

A Double-edged notched test (DENT) whose specimen geometry is depicted in Fig. 3a is assessed. The phase field problem (13) is solved using two procedures: (a) the compact history function, as proposed in Eqs. (14–16), and (b) the Bound-Constrained Optimization Problem (BCOP) methodology, as proposed in equation (18).

Uniform vertical displacements in the top and bottom edges of the specimen are prescribed. Assuming two symmetric cracks propagating from both notches, only one-fourth of the specimen is modeled. The model parameters are Young modulus $E = 2.5$ GPa, Poisson's ratio $\nu = 0.25$, fracture energy $G_c = 5$ N/mm, and characteristic length: $\ell = 1.8$ mm.

A uniform finite element mesh, with element sizes $1.67h_e \times h_e$ ($h_e = 0.1$ mm), is taken for modeling the crack propagation region.

The accuracy of solutions is estimated by computing the area, $W^{ext} = \int_0^\infty 2f(\delta)d\delta$, enclosed by the force-displacement curve once the specimen reaches the complete load capacity loss. Under linear elastic fracture conditions, the area should reproduce the energy release W^E (per unit thickness) ruled by the fracture energy G_c :

$$W^E = G_c \frac{b}{2}, \quad (50)$$

where the term $b/2$ is the crack path length at the end of the crack propagation process. Thus, the relative percent error of the energy release is:

$$e_W = \frac{W^{ext} - W^E}{W^E} \times 100. \quad (51)$$

4.2.1 Results obtained with the compact history function methodology

Results of the numerical assessments, in terms of force $f/2$ versus displacement δ , are plotted in Fig. 3b. We show three results run with phase field enrichment at the crack tip (PFTE) and different tolerances $\Delta\hat{\phi}^{opt}$ (10^{-2} , 5×10^{-3} , and 2.5×10^{-3} , respectively). We also show the solution attained without phase field enrichment (without PFTE) and $\Delta\hat{\phi}^{opt} = 10^{-2}$. The Figure inset depicts a zoom of the structural limit load region. The limit loads of solutions obtained with and without PFTE are notably different, a feature that agrees with the results reported in the literature ([18], [24]).

Though the results with PFTE look very similar in all three cases, the accuracy measured in terms of the error, e_W , depicted in Table 1, gives a different insight.

The mechanical partition of the staggered scheme converges to the required tolerance in all steps, independently of the parameter $\Delta\hat{\phi}^{opt}$. Convergence of the mechanical partition demanded between 3 to 7 iterations per step.

The plot in Fig. 3c displays the evolution of the CTOD (Crack Tip Opening Displacement) measured at the original crack tip versus the step number. This variable shows a non-monotonous increase as the crack propagates. Therefore, due to the non-monotonic behaviour, the CTOD is not an adequate variable that can be taken as the arc-length parameter.

Identifying the crack tip as the point on the symmetry x -axis intersected by the phase field iso-level curve of value $\varphi = 0.92$, then, Fig. 3d plots the crack length versus the step number. Notice that the arc-length control strategy strictly regulates the crack tip advance during the snap-back process.

4.2.2 Results obtained with the BCOP methodology.

The effectiveness of the arc-length control strategy is additionally examined by solving the damage stage of the staggered scheme through the BCOP methodology defined with Eq. (18). This BCOP methodology is implemented with the interior-point-convex algorithm of Matlab.⁴ The mechanical stage is identical to that used with the compact history function approach, i.e., the Eq. (23) and the control Eq. (27) are solved together in this stage.

Figure 4a compares the load versus displacement curves obtained with the BCOP methodology and the compact his-

⁴ Quadprog Matlab routine.

Fig. 3 Double-edged notched test (DENT). **a** Specimen geometry. **b** Half of the total force f versus displacement δ . Solutions obtained with $\Delta\hat{\phi}^{opt}$: 10^{-2} , 5×10^{-3} , and 2.5×10^{-3} (with PFTE) and 10^{-2} (without PFTE). **c** CTOD versus step number. **d** Crack tip position versus step number

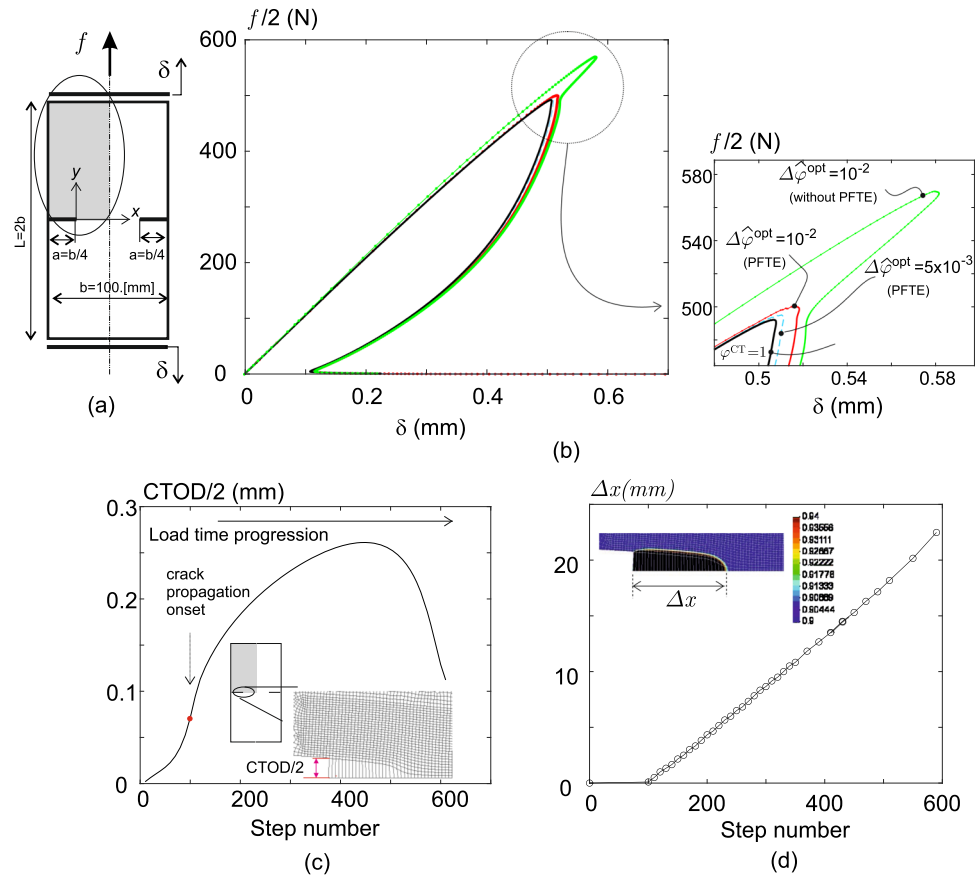


Table 1 Double-edged notched test (DENT)

$\Delta\hat{\phi}^{opt} (\times 10^{-2})$	Compact history function		BCOP methodology	
	e_W	Number of steps	e_W	Number of steps
2.5	—	—	24%	220
1.0	13%	600	12.3%	585
0.5	10%	1150	8.5%	1300
0.25	8%	2340	6.7%	2450

Relative percent errors and number of steps required for different runs. Solutions obtained with PFTE. Comparison between procedures using compact history function and BCOP methodologies

tory function with tolerance $\Delta\hat{\phi}^{opt} = 0.5 \times 10^{-2}$ and PFTE. Both solutions are almost indistinguishable. However, Fig. 4b displays notable differences in phase field distributions along the crack path transverse direction. The phase field obtained with the compact history function shows a more dispersed solution than that obtained with the BCOP. This feature agrees with similar comparisons reported in the literature.

Table 1 compares the errors, e_W , defined in Eq. (51) in different runs. It also reports the number of steps to attain the displacements $\delta = 0.2$ mm for the BCOP solution and $\delta = 0.7$ mm for the compact history function solution. In this sense, the performance is similar for both methodologies. However, the problem solved with the BCOP methodology demands a substantially higher CPU time than that required by the compact history function methodology.

Figure 11 in Sect. 5 examines additional outcomes of the numerical efficiency of the arc-length control strategy combined with the BCOP approach for solving this test.

4.3 Notched plate with hole (NPWH)

The notched plate with an eccentric hole (NPWH) depicted in Fig. 5a is evaluated. The load consists of a prescribed displacement δ . This plate has been experimentally tested, numerically assessed, and reported in the work of Ambati et al. [32].

The material used to perform the experimental test is concrete. However, in the present simulation, we adopt similar model parameters to those taken in the reference work, i.e., Young modulus $E = 5.98$ GPa, Poisson's ratio $\nu = 0.221$

Fig. 4 Double-edged notched test (DENT). Comparison of results obtained with the compact history function and BCOP approaches. **a** Half of the total force f versus displacement δ , tolerance $\Delta\hat{\varphi}^{opt}: 0.5 \times 10^{-2}$ (with PFTE). **c** Phase field along a line orthogonal to the crack path, at the end of analysis

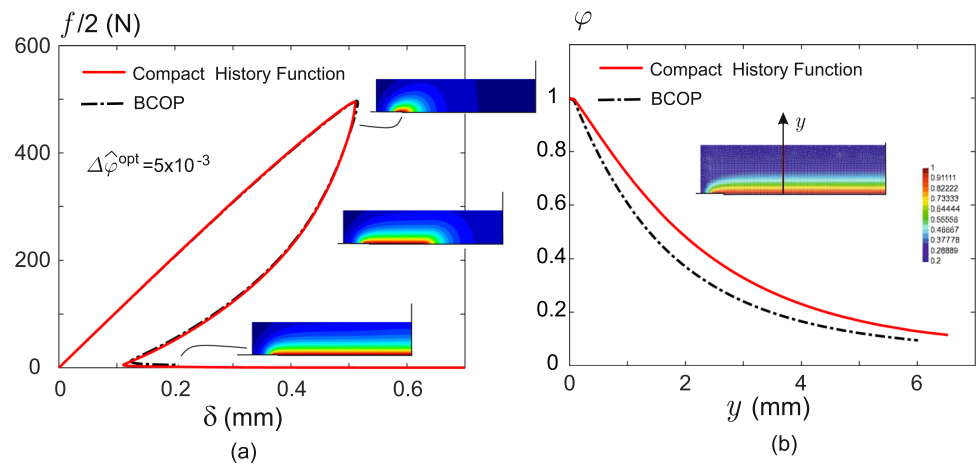
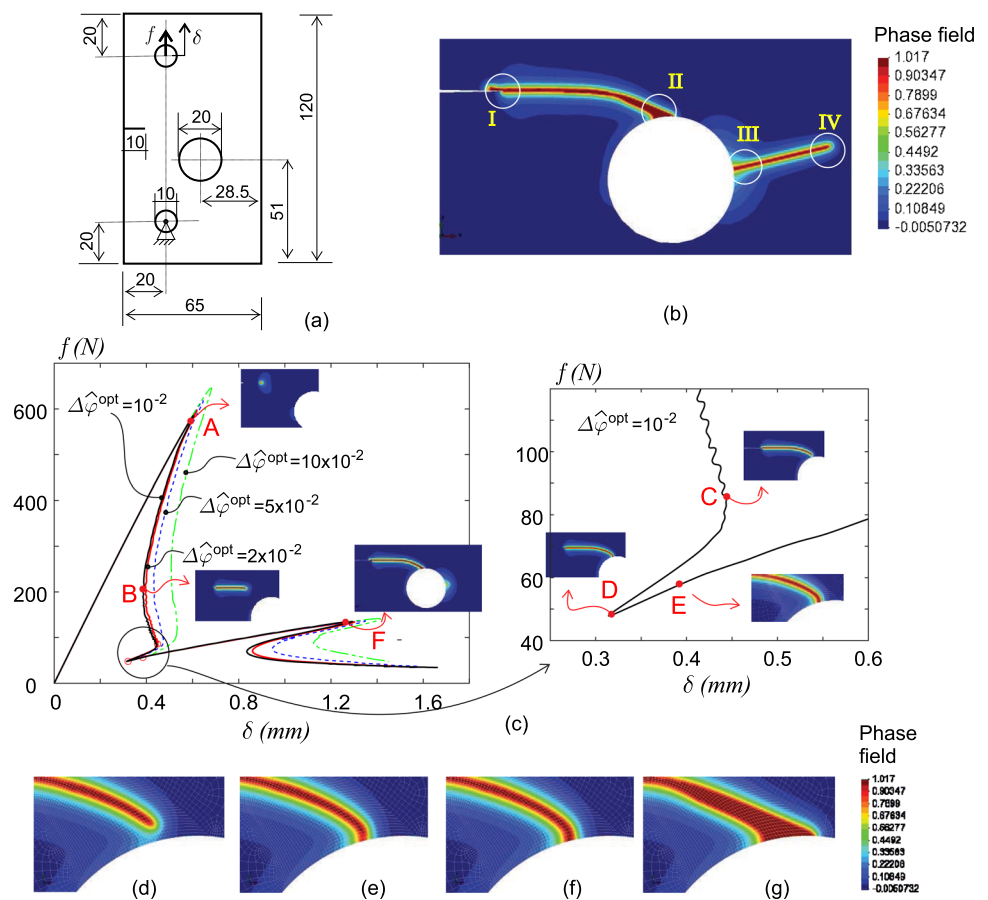


Fig. 5 Notched plate with a hole (NPWH). **a** Specimen geometry. **b** Crack path at the end of the analysis ($\delta = 2$ mm). **c** Force f versus displacement δ . **d–g** Sequence of crack advance images when the crack intersects the hole. Images: **d–g** correspond to Points C, D, E, and F in the equilibrium curve, respectively



and $G_c = 2.28$ N/mm. The phase field characteristic length is $\ell = 0.5$ mm, which is five times higher than that adopted by Ambati et al. The value $\ell = 0.5$ mm allows us to use a coarse mesh with 8500 bilinear quadrilateral finite elements. In the region of the specimen where the crack is expected to propagate, the mesh is uniform with element sizes between 0.15 mm and 0.25 mm. Tests are run without PFTE and $\Delta s^{max} = 0.6$ Nmm.

4.3.1 Results

The attained crack path at the end of the analysis, when $\delta = 2$ mm, is shown in Fig. 5b. It is similar to that observed in the experimental and numerical results of the reference work. The plots of load f versus displacement δ curves obtained with tolerance $\Delta\hat{\varphi}^{opt} = 10^{-2}$, 2×10^{-2} , 5×10^{-2} , and 10×10^{-2} , are shown in Fig. 5c. The inset of this Figure depicts a

Table 2 NPWH specimen

$\Delta\hat{\varphi}^{opt} (\times 10^{-2})$	e_W (%)	Number of steps
2	3	1730
5	16	665
10	45	280

Relative percent error for different runs and number of steps (staggered strategy)

zoom of these plots in the snap-back region, coinciding with the instant when the crack reaches the eccentric hole.

The errors of the solutions in different runs are assessed with expression (51). Here, W^E is the area enclosed by the curve with tolerance $\Delta\hat{\varphi}^{opt} = 10^{-2}$. This solution is taken as the reference one. These errors are shown in Table 2.

The initial crack starts at the sharp notch root (Point I in Fig. 5b) in coincidence with point A in the load–displacement curve. Then, it propagates until its intersection with the eccentric hole at point II in Fig. 5b. The equilibrium curve displays a marked snap-back effect when the crack is close to intersecting the hole boundary, coinciding with Point C in the Figure inset, details of the crack morphology are shown in Fig. 5d. The loading process continues until reaching Point D, details are shown in Fig. 5e. As expected, the crack deviates its direction to intersect the hole boundary orthogonally, such as required by the boundary condition in Eq. (5)-b. A posterior reload, from Point D to F, is observed. The parameter Δs^{max} plays a role in this loading stage by limiting the maximum step length. During this reloading process, in the neighborhood of point II shown in Fig. 5b, there is a stress state with positive principal stresses parallel to the crack. This effect induces a stable damage increase and a widening of the area with $\varphi = 1$, as shown in Fig. 5g. The features observed in the sequence of pictures in Fig. 5d, e, f, and g are captured with a precise crack advance control strategy supplied by the arc-length technique.

Also, the reloading process generates a second crack starting on the smooth boundary at the right part of the eccentric hole. This instant corresponds to Point F in Fig. 5c. The propagation process of this second crack, from Point III to Point IV in Fig. 5b, describes an equilibrium curve with a well-marked snap-back effect.

The characteristic equilibrium curve of this test displays a similar feature to that shown by the specimen with an eccentric hole in Fig. 8. Two limit points define the nucleation of both cracks, and two well-marked snap-backs determine the equilibrium curve at posterior loading regimes. However, in this test, the nucleation of both cracks has different natures. The material fracture energy governs the crack nucleation in the notch root. Instead, the material strength governs the crack nucleation on the right part of the eccentric hole; the lack of a pre-existing notch produces this effect. Hence, the

first crack is not susceptible to the material strength, and consequently, it is not sensitive to the characteristic length parameter ℓ . By which, the limit load (Point A in Fig. 5c) of our numerical result agrees well with the solution of Ambati et al. Alternatively, the second crack is sensitive to the material strength and governed by the parameter ℓ . So, the corresponding limit load, point F in Fig. 5c, does not coincide with that reported by Ambati et al. Further details of the discussion about crack nucleation in both cases can be seen in the works of Kumar et al. [33].

4.4 Single edge notched shear test (SENST)

The (SENST) square specimen with size 1×1 mm shown in Fig. 6a is evaluated. This specimen undergoes prescribed displacements on its boundary compatible with a pure shear deformation mode. A phase field approach assesses the fracture path with model parameters $E = 210$ GPa, $\nu = 0.3$, $G_c = 2.7$ N/mm, and characteristic length $\ell = 0.015$ mm. This test has been widely solved and reported in the literature using a phase field approach. In Sect. 5, we compare the performance of these results with our assessments.

We evaluate the equilibrium curve using two finite element meshes: a) the coarse mesh with 13,100 bilinear quadrilateral elements is homogeneous only in the region intersected by the crack, and the finite element size in this zone is $h^{ef} \approx \ell/4$, b) the finer mesh with 390,990 bilinear quadrilateral elements is a homogeneous structured mesh in the complete domain, with finite element sizes $h^{ef} \approx 0.0016$ mm. Phase field tip enrichment (PFTE) is modeled in all cases. Using the coarse mesh, we obtain four equilibrium curves with tolerances $\Delta\hat{\varphi}^{opt}$, 0.5×10^{-2} , 1×10^{-2} , 2×10^{-2} , and 4×10^{-2} . The force F versus displacement δ curves are shown in Fig. 6b. The structural behaviour in these cases is stable under prescribed displacements, except when the crack reaches the bottom part of the specimen where snap-backs are captured. The subsequent structural responses, from Point A to B, display a spurious apparent strength increase attributed to the combined effect of compressive stresses in the lower right part of the specimen and the adopted type of free energy partition defined in Eq. (8).

Figure 6c compares F versus δ solutions obtained with the fine and coarse meshes and $\Delta\hat{\varphi}^{opt} = 4 \times 10^{-2}$ in both cases.

Accuracy versus computational cost assessment. Table 3 shows the errors of our coarse mesh numerical solutions assessed with the expression (51). W^E is the area enclosed by the solution with tolerance $\Delta\hat{\varphi}^{opt} = 0.5 \times 10^{-2}$ and coarse mesh taken as the reference one. The Table also describes the number of steps required for solving the staggered scheme until reaching the displacement $\delta = 0.014$ mm. The com-

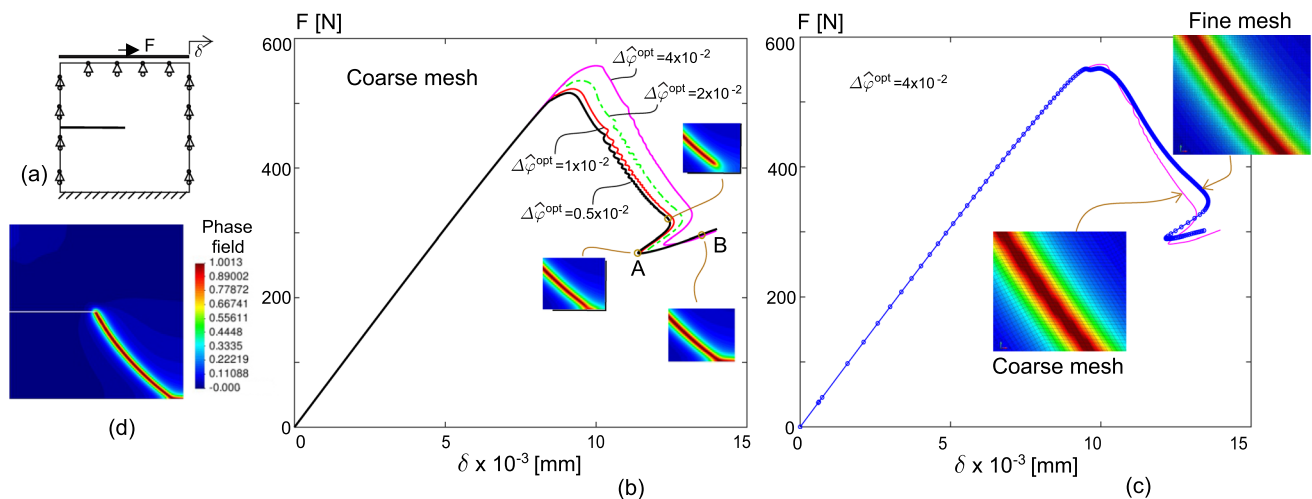


Fig. 6 Single edge notched shear test (SENST). **a** Specimen geometry and boundary conditions. **b** Coarse mesh: total shear force F on the top boundary versus displacement δ . The insets show the phase field variable evolution in three instants as the snap-back develops along the

equilibrium curve. Phase field tip enrichment (PFTE) has been assumed. **c** F versus δ curves, coarse and fine meshes with $\Delta \hat{\phi}^{opt} = 4 \times 10^{-2}$. Points of the fine mesh result indicate load steps. **d** Crack path at the end of the analysis ($\delta = 0.014$ mm)

Table 3 SENST specimen

$\Delta \hat{\phi}^{opt} (\times 10^{-2})$	e_W	Number of steps	CPU time (sec)
0.5	—	3540	5907
1	1.4%	1697	2249
2	4.2%	826	990
4	9.2%	394	451

Relative percent error for different runs, number of steps (staggered strategy), and CPU times

putational cost, CPU time,⁵ demanded in every run is also reported in the last column.

Additional discussion on the performance of the staggered scheme combined with our arc-length strategy for solving this problem are analysed in Sect. 5.

4.5 Fracture of an adhesively-bonded composite

The strength of a specimen composed of two plates bonded with an adhesive interface is evaluated. Figure 7a shows the specimen and the applied load. The expected fracture process in this problem results from two interacting mechanisms, interface delamination followed by a possible kinking out of the interface crack. The position d where the crack branches out the interface, see Fig. 7k, results from complex inter-

acting factors, some of which have been studied by Li et al. [34]. Interestingly, the crack branching out phenomenon depends not only on the relative plate/interface toughness, as explained from a Linear Elastic Fracture Mechanic perspective, but on the relative strengths of the plate and interface.

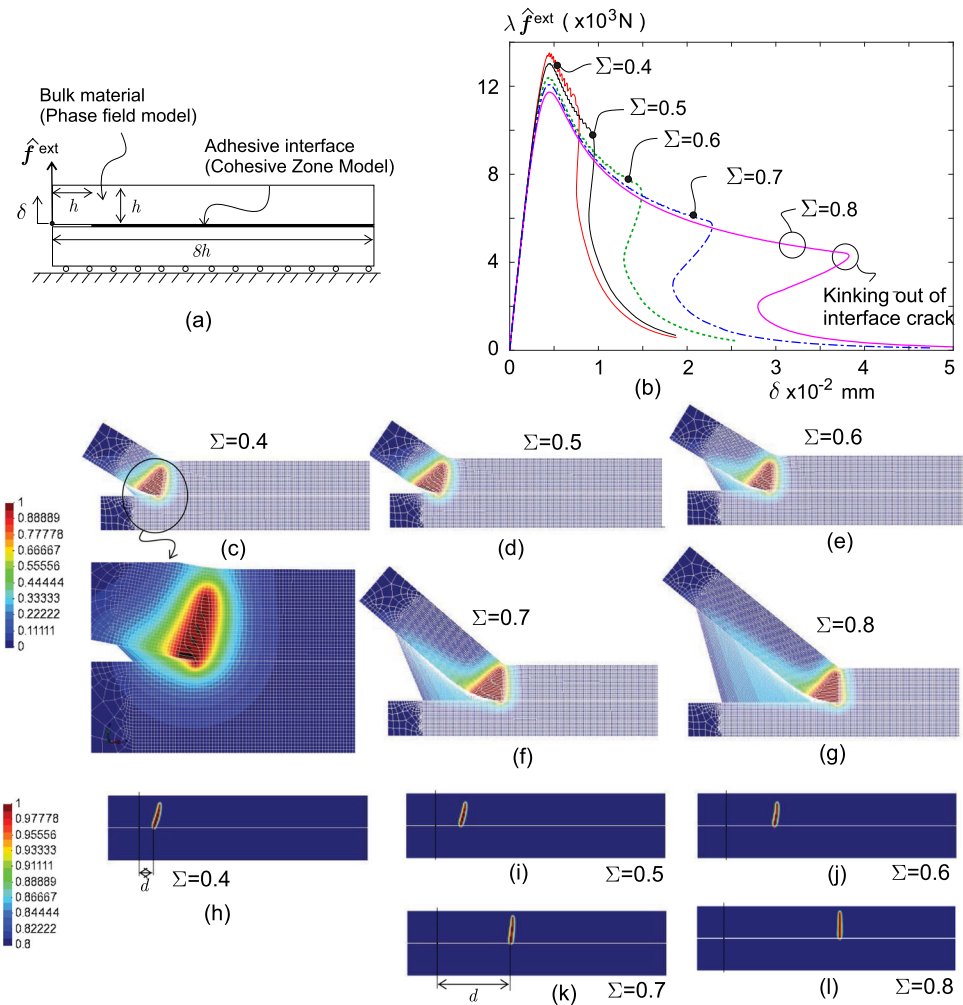
The mechanical model for simulating this problem is adopted from [35]. In this approach, the phase field model is combined with cohesive interfaces characterized by a conventional traction-separation law not influenced by the phase field variable. The phase field variable and the micro-stress component, which is normal to the interface, are constrained to be continuous across the interface. As mentioned above, the characteristic length of the phase field model in this test takes the role of a material parameter.

This test aims to evaluate the effectiveness of the arc-length control strategy in managing fracture problems undergoing distinct degradation mechanisms, such as interface decohesion and propagating cracks in the plate modeled with different techniques. Both of these mechanisms can trigger loading process instabilities. Therefore, the arc-length procedure needs to handle them both. Note that arc-length strategies controlling the phase field variable are not adequate in this case.

As mentioned before, the mechanical stage of the staggered procedure solves the displacements of a problem with fixed phase field variables. This stage takes into account the non-linearities associated with the interface degradation modeling. Therefore, the control Eq. (27) is here extended to manage the instabilities triggered by the interface decohesion. We define a control equation regulating the rate of the driving force term of the micro-force phase field equation as

⁵ Our results are obtained with a desktop computer, with an AMD Ryzen 9 5900X 12-Cores processor, and a home-made Matlab code using sparse matrices and “parfor” for assembling global matrices. Running the same problems in sequential mode, with only one core, the CPU times for the instances $\Delta \hat{\phi}^{opt} = 1 \times 10^{-2}$ and $\Delta \hat{\phi}^{opt} = 4 \times 10^{-2}$ have been 9975.sec and 2850.sec, respectively.

Fig. 7 Adhesively-bonded composite. **a** Specimen geometry and load. **b**) Force $\lambda \hat{\mathbf{f}}^{ext}$ versus displacement δ . **c–g** deformed meshes and phase field distribution at the end of the analysis. **h–l** crack path in the plate at the end of the analysis



well as the power dissipation due to the interface degradation, $\mathcal{D}_{CZM}$. A convex combination of both terms is taken as follows:

$$(1 - \xi) \int_{\mathcal{B}} 2(1 - \varphi) \dot{\mathcal{H}} d\mathcal{B} + \xi \mathcal{D}_{CZM} = \dot{s}, \quad \xi \in [0, 1]. \quad (52)$$

Following [17], the rate of dissipation $\mathcal{D}_{CZM}$ in the body $\mathcal{B}$ undergoing the degradation of the interface cohesive forces, and which are simulated with a damage model, can be evaluated through the expression:

$$\mathcal{D}_{CZM} = \frac{1}{2} \left(\lambda(\hat{\mathbf{f}}^{ext} \cdot \dot{\mathbf{u}}) - \dot{\lambda}(\hat{\mathbf{f}}^{ext} \cdot \mathbf{u}) \right). \quad (53)$$

Considering (52) and (53), the control equation expressed in increments is now rewritten as follows:

$$(1 - \xi) \int_{\mathcal{B}} 2(1 - \varphi) \Delta \mathcal{H} d\mathcal{B} + \frac{\xi}{2} \left(\lambda(\hat{\mathbf{f}}^{ext} \cdot \Delta \mathbf{u}) - \Delta \lambda(\hat{\mathbf{f}}^{ext} \cdot \mathbf{u}) \right) - \Delta s = -res_s, \quad (54)$$

where λ and $\mathbf{u}$ take the values at the end of the previous step of the staggered scheme. Finally, after replacing Eqs. (35) and (36) into (54) and operating on the algebraic expression, we determine the formulas for updating $\Delta \lambda$ and $\Delta \mathbf{u}$ that substitutes the Eq. (37) in the control strategy algorithm.

Geometric and material parameters. The plate thickness is $h = 2.5 \text{ mm}$, and its elastic modulus: $E_p = 100 \text{ MPa}$, fracture energy $G_p = 2.5 \times 10^{-4} \text{ N/mm}$, and characteristic length $\ell_c = 0.06 \text{ mm}$. Thus, the plate critical stress is estimated through expression (46), $\sigma_p^{crit} = 2.1 \times 10^{-1} \text{ MPa}$. The adhesive interface follows a damage law with exponential softening characterized by the fracture energy $G_i = 1.5 \times 10^{-4} \text{ N/mm}$, an elastic modulus E_i assumed identical to that of the plate, $E_i = E_p$, and the interface strength σ_i

is given in terms of the parameter Σ defined as the stress ratio: $\Sigma = \sigma_p^{crit} / \sigma_i$. Then, the shortest interface characteristic length corresponds to the case $\Sigma = 0.4$, and its value is estimated as $\ell_i = 0.53 E_i G_i / (\sigma_i)^2 = 0.0289$ mm.

Two uniform structured quadrilateral finite element meshes of sizes $h^{EF} = 0.0175$ mm and $h^{EF} = 0.00875$ mm are used for the numerical evaluations.

4.5.1 Parametric assessment of the composite fracture problem

Failure of the specimens is studied through a parametric analysis by varying Σ . Five cases with $\Sigma = 0.4, 0.5, 0.6, 0.7$, and 0.8 and using the arc-length control strategy are solved. Figure 7b displays the force versus displacement plots. Additionally, the fracture modes attained with this methodology are depicted in Fig. 7c–g, and the crack paths along the plate are shown in Fig. 7h–l, respectively. A deformed mesh detail of the case with $\Sigma = 0.4$ is also shown in the same Figure.

Considering that the test is undergoing a prescribed force, the arc-length strategy plays a role in tracing the equilibrium curves across the limit loads as well as in loading stages coinciding with the instant when the interface crack kinks out toward the plate. Responses displaying snap-backs are captured after the crack kinks out the interface. This feature is more pronounced as the parameter Σ is higher. In cases characterized with low values of Σ , i.e., 0.4 and 0.5 , the control strategy responds adequately by assuming $\xi = 0.5$, meaning that a relatively similar weight is given to both, the phase field driving force term and the interface dissipation rate, in the control equation. Alternatively, in cases with high values of Σ , it is more convenient to adopt $\xi = 0.25$ that assigns a relatively high weight to the phase field driving force term in the control equation.

We evaluate the distance d at which the crack kinks out the interface using both finite element meshes. These results are

Table 4 Fracture of an adhesively-bonded composite

Σ	Fine mesh ($h^{EF} = 0.00875$ mm)	Coarse mesh ($h^{EF} = 0.0175$ mm)
	d/h	d/h
0.4	0.4	0.42
0.5	0.66	0.84
0.6	1.36	1.72
0.7	2.28	2.64
0.8	3.4	4

Distance d is assessed with two finite element meshes

displayed in Table 4. Notice that slightly smaller distances d are estimated with the fine mesh.

4.6 Plate with an eccentric hole in tensile (EH)

The square plate with an eccentric hole (EH) undergoing vertical stretching, shown in Fig. 8a, is evaluated. A uniform vertical displacement, δ , is prescribed on the top edge of the specimen. The specimen does not have pre-established notches. Therefore, localization of damage in the phase field model is governed by a strength criterion based on the characteristic critical stress σ^{crit} defined in (44) (for $\varphi^{crit} = 0.25$).

We adopt material properties similar to those reported by Bharali et al. [19] for solving the same problem. The Young modulus is $E = 210$ GPa, Poisson's ratio $\nu = 0.3$, $G_c = 2.7$ N/mm, and $\ell = 0.02$ mm ($\sigma^{crit} = 1.73$ GPa). The mesh has 4652 quadrilateral elements.

The equilibrium solution in terms of the load f versus displacement δ displays a characteristic behaviour with two limit points followed by two snap-backs, as shown in Fig. 8b. The first snap-back initiates when the crack propagation starts on the specimen at the right part of the eccentric hole. The second snap-back initiates when the second crack starts its propagation on the left part of the eccentric hole. Four solutions are

Fig. 8 Plate with an eccentric hole (EH). **a** Specimen geometry. **b** Total force f versus displacement δ

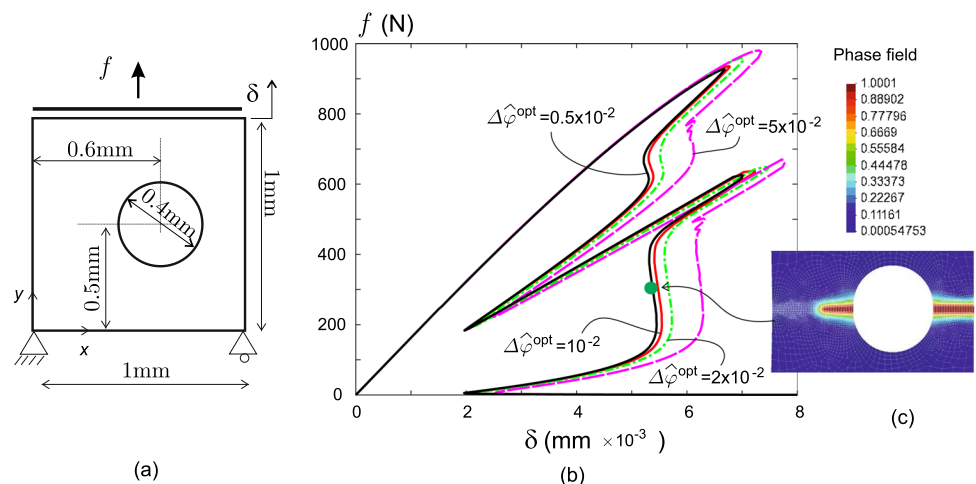


Table 5 EH specimen

$\Delta\hat{\varphi}^{opt} (\times 10^{-2})$	e_W	Number of steps
1	3.7%	1241
2	10.8%	610
5	31.3%	244

Relative percent error for different runs and number of steps (staggered strategy)

plotted with the control parameters $\Delta\hat{\varphi}^{opt} = 0.5 \times 10^{-2}$, 1×10^{-2} , 2×10^{-2} , and 5×10^{-2} . The errors of these solutions, e_W , are assessed with equation (51), where the areas enclosed by the curves, W^{ext} , are compared with the area enclosed by the reference solution, W^E that is the curve obtained with $\Delta\hat{\varphi}^{opt} = 0.5 \times 10^{-2}$. Table 5 show the errors and the number of steps required for the staggered strategy in every run.

The maximum arc-length parameter is $\Delta s^{max} = 0.008$ Nmm. Additional computational cost performances of this test are discussed in Sect. 5.

4.7 Perforated plate

We simulate the perforated plate shown in Fig. 9, which is undergoing a loading process under the forces f . This test is reported in May et al. [29] from where we adopt the same elastic model parameters. The Young modulus is $E = 100$ MPa and Poisson's ratio $\nu = 0.3$. In the reference work, the material degradation is modeled with interface elements along the x-axis. The fracture energy of the interface model is $G_c = 2.5 \times 10^{-3}$ N/mm, and the ultimate stress is $\sigma^{ult} = 1$ MPa. In the present phase field model, we preserve

the same G_c whereas ℓ is defined such that the critical stress, σ^{crit} , determined by expression (44), coincides with σ^{ult} of the interface model. The so-determined characteristic length is $\ell = 0.0812$ mm. We use a mesh with 2396 quadrilateral elements.

Figure 9b plots the force f versus displacement δ . We show two solutions with $\Delta\hat{\varphi}^{opt} = 10^{-3}$ and 5×10^{-3} , respectively. Both solutions require approximately 600 and 1200 steps. All steps converge. The limit loads in this plot agree with those of the reference work.

The maximum arc-length parameter is $\Delta s^{max} = 6 \times 10^{-6}$ Nmm, which dominates the control strategy during the unloading steps after crossing the peak loads.

5 Performance analysis of the staggered scheme combined with the arc-length strategy

Further performance on the computational costs for solving the EH, NPWH, and SENST tests discussed in sub-Sections 4.6, 4.3, and 4.4 are analyzed. A summary of these analyses is described in Figs. 10. Additionally, Fig. 11 displays the performances of the DENT numerical test attained with the BCOP methodology in sub-Sect. 4.2.

The plots on the left column of Figs. 10 and 11 depict the computed phase field increment $\Delta\varphi = \|\varphi_n - \varphi_{n-1}\|_\infty$ in every step of the staggered scheme using the arc-length strategy. Solutions for different values of $\Delta\hat{\varphi}^{opt}$ are shown. From Sect. 3.2.1, Eq. (38), we recall that our strategy adjusts the arc-length parameter, s , to attain increments $\Delta\varphi$ as close as possible to an optimum desired phase field increment $\Delta\hat{\varphi}^{opt}$.

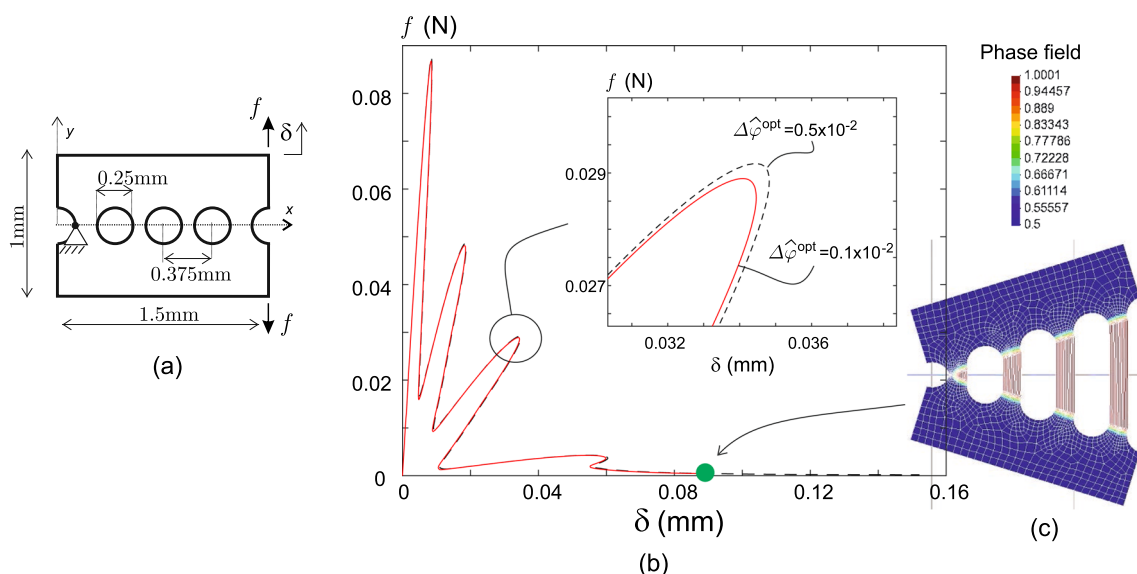
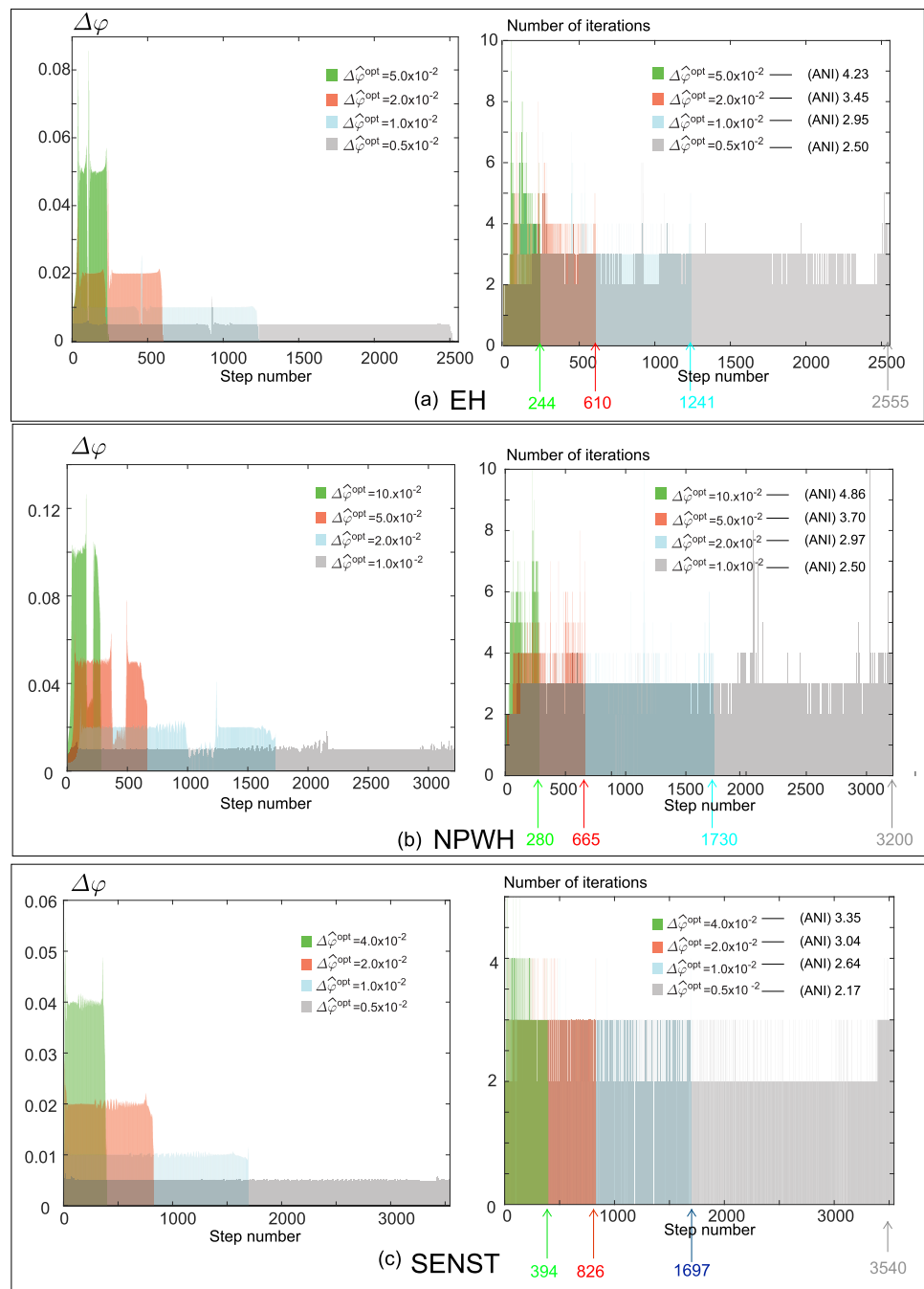


Fig. 9 Perforated plate. **a** Specimen geometry. **b** Force f versus displacement δ curves

Fig. 10 Performance analysis. **a** EH, **b** NPWH, and **c** SENST test. Left column: phase field increase ($\Delta\varphi = \|(\varphi_n - \varphi_{n-1})\|_\infty$) in each staggered scheme step (Step Number). Right column: number of iterations required by the mechanical stage solution in each staggered scheme step. Load versus displacement plots for the here reported tests, EH, NPWH, and SENST, are shown in Figs. 8b, 5c, and 6b, respectively



Fulfillment of this condition means that the pseudo-time integration error holds bounded. In the four studied cases, we note that this characteristic feature of the algorithm is satisfied with a high degree of accuracy independently of the algorithm imposing the complementarity conditions in the micro-force balance equation.

The plots on the right column of both Figures display the number of Newton Raphson iterations (*niter*) necessary for solving the mechanical stage per each staggered scheme step (*nstep*). This information sheds light on the computa-

tional cost of every run. One staggered step of the present explicit approach requires one mechanical and one phase field stage evaluation per step. Also, one mechanical stage solution demands *niter* Newton Raphson iterations. So, the total number of internal force vector and stiffness matrix evaluations is *niter* + 1. In the plots, we also depict for every run the average number of iterations per step (ANI). Therefore, the estimated number of times that the (mechanical) internal force vectors and stiffness matrices are computed in one run is $(ANI + 1) \times nstep$, while the estimated number of times that

Fig. 11 Performance analysis. DENT test using the Bound-Constrained Optimization Problem (BCOP) methodology for solving the micro-force balance equation. Left column: phase field increase ($\Delta\varphi = \|\varphi_n - \varphi_{n-1}\|_\infty$) in each staggered scheme step (Step Number). Right column: number of iterations required by the mechanical stage solution in each staggered scheme step

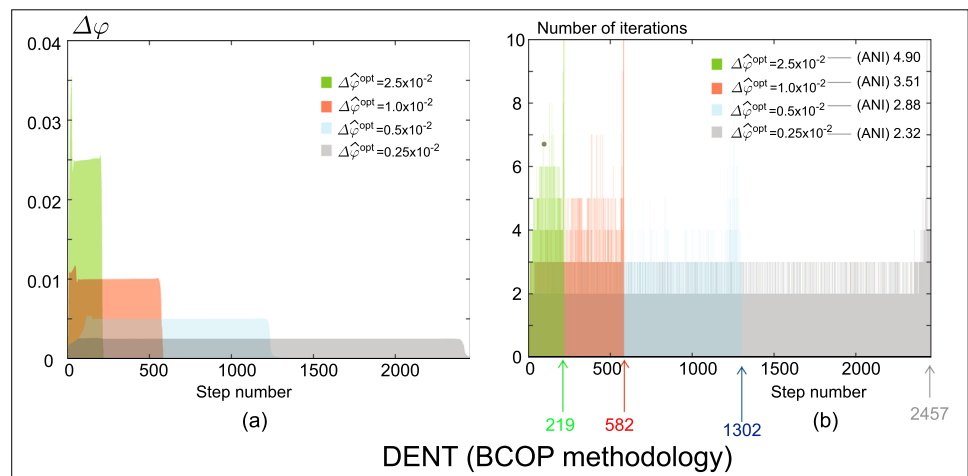


Table 6 Performance comparative analysis using the SENST test

Reference paper	Methodology	Mesh (N - FE)	nsteps	nsteps × niter AVG	Capture snap-backs?	CPU time (sec)
[36]	M (BFGS)	390,000 FE	100	3131	NO	60,780
[36]	STG	390,000 FE	100	9374	NO	417,600
[3]	STG	11,200 N	46	≈ 100 (1)	NO	6840 (2)
			staggered	iter/		59,760 (3)
			step	stagg. step		
[3]	M	11,200 N	46	≈ 30	NO	10,800
	(with lineasarch)			iter/		
				stagg. step		
[12]	STG	16,384 FE	135	30 (4)	NO	NR
[19]	M (arc-length + linesarch)	Adaptative	127 (5)	1966	YES	2097
Present	STG	13100FE	394	1319	YES	451 (6)
	STG	13100FE	1697	4480	YES	2249 (7)
	STG	390990FE	420	1596	YES	17,760 (8)

M, monolithic; STG, staggered; nstep, number of monolithic/staggered steps; niter AVG, average number of Newton–Raphson iterations required for solving one mechanical stage. NR, not reported; N, nodes; FE, finite elements. (1) the number of Newton–Raphson iterations required per each mechanical stage solution is not reported. (2) CPU-time for coarse tolerance (10^{-2}). (3) CPU-time for adjusted tolerance (10^{-4}). (4) the Maximum number of iterations for solving a staggered step is prescribed to 30. No information is reported on the number of Newton Raphson iterations required for solving the mechanical stage. The number of staggered steps is not specifically reported. (5) Solved steps: 66, Refined steps: 50, Failed steps: 11. (6) CPU time for run with coarse mesh and $\Delta\hat{\varphi}^{opt} = 0.04$ ($e_w = 9.2\%$). (7) CPU time for run with coarse mesh and $\Delta\hat{\varphi}^{opt} = 0.01$ ($e_w = 1.4\%$). (8) CPU time for run with fine mesh and $\Delta\hat{\varphi}^{opt} = 0.04$

the phase field internal force vectors and the corresponding stiffness matrices are computed in one run is nstep. The number of steps required for solving different runs in these plots is indicated in color below the x-axis. This information, jointly with the equilibrium curves and accuracy analysis depicted in previous Sections, provides an insight into the expected accuracy in terms of the required computational cost for different runs.

Considering that the SENST problem is a widely used benchmark to assess the performance of various phase field algorithms, we compare the performance of our approach with that reported in the literature. Table 6 compares several

aspects of this analysis. Based on these results, we conclude that our methodology entails a moderate computational cost and provides solutions exhibiting snap-back behaviour even when using a large arc-length parameter, $\Delta\hat{\varphi}^{opt}$.

6 Conclusions

An arc-length procedure for controlling brittle fracture problems modeled with a phase field approach, whose balance equations are solved with a well-established staggered strategy, is proposed. The new contribution in this paper refers

to the control equation governing the increment of displacements solved in the mechanical stage of the staggered strategy. This control equation determines, through an arc-length parameter, s , the driving force increase defining the phase field increase via the micro-force balance equation.

Compared with previously reported arc-length strategies, the control equation presented in this work allows for solving a wide range of problems, including those with the micro-force balance equation evaluated through optimization techniques with inequality constraints or those with multiple instability mechanisms.

The arc-length procedure implementation demands the evaluation of an additional internal force vector during the mechanical stage that can be calculated in combination with the conventional internal force vector. Also, similar to traditional arc-length procedures, a double back substitution is needed to evaluate $\Delta\lambda$ and $\Delta\mathbf{u}$ in each iteration of the mechanical stage. According to our numerical experiments, these operations do not significantly increase the computational burden of the original staggered strategy.

Through the numerical assessments discussed in Sect. 4, we show the robustness, accuracy, and ability of the proposed arc-length strategy to trace equilibrium curves displaying abrupt snap-back responses, and to control quasi-static crack advance problems that, under displacement or force control techniques, are unstable. The crack advance is strictly controlled in every step, regardless of the structural brittleness of the assessed problem.

Accuracies of the numerical solutions in Sect. 5 have been assessed through the area enclosed by the equilibrium curves. Considering brittle fracture problems and fully degraded notched specimens, the enclosed area of the force versus displacement equilibrium curve identifies the energy release in the test, which should be identical to the specimen fracture energy times the crack length. This criterion can be used as a strict form of verifying the energetic consistency of numerical solutions. Numerical methodologies which cannot trace the complete equilibrium curves, including snap-backs, do not pass this verification test.

The arc-length parameter is adjusted to yield an approximate pre-established desired damage increment per step. This feature is implemented through the $\Delta\hat{\phi}^{opt}$ parameter. Then, the damage increase in each step stays controlled, even using an explicit staggered strategy. According to results presented in Sect. 5, this procedure entails moderate computational costs for tracing the complete equilibrium curve, with the subsidiary advantage that we have never detected lack of convergence in the tests presented in Sect. 4, and it is not necessary to use line search techniques.

Acknowledgements This research has been supported with grants provided by CONICET: PIP 112202101 00455CO (2022–2024), the National University of Litoral, and the fund from ANPCyT: PICT-

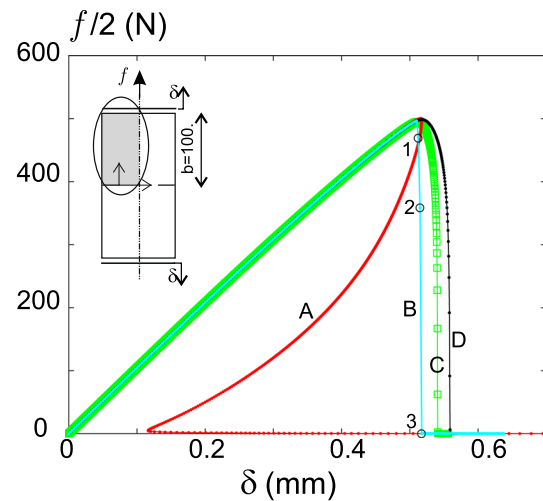


Fig. 12 Double-edged notched test (DENT). Solutions obtained with: arc-length procedure (curve A); variant I and $\Delta\delta/b = 25. \times 10^{-6}$ (curve B); variant II and $\Delta\delta/b = 2.5 \times 10^{-6}$ (curve C); variant II and $\Delta\delta/b = 5. \times 10^{-6}$ (curve D)

Table 7 Computational cost for solving the DENT test defined in terms of $nstep \times n_{subiter} \times niter$, where $nsteps$ is the number of load steps, $n_{subiter}$ is the number of sub-iterations in variant I, or 1 in variant II, and $niter$ is the number of Newton–Raphson iterations required by the mechanical stage

Staggered strategy	Step length	$nsteps$ $n_{subiter}$ $niter$	$\times$ $\times$	$e_W(\%)$
Variant I	$\Delta\delta/b = 25. \times 10^{-6}$	2403		116
Variant II	$\Delta\delta/b = 2.5 \times 10^{-6}$	3271		133
Variant II	$\Delta\delta/b = 5 \times 10^{-6}$	2265		147
with arc-length	$\Delta\hat{\phi}^{opt} = 1. \times 10^{-2}$	1860		13

Last column is the relative percent error of the energy release, e_W , defined in Eq. (51)

2020-SERIEA-02793, Argentina. F.P. Duda gratefully acknowledges the support provided by the Brazilian agency CNPq (309370/2019-1).

Appendix I: Solutions obtained with staggered strategies without using the arc length procedure

This appendix describes supplementary solutions for the DENT test subjected to uniform stretching studied in Sect. 4.2. Two variants of the staggered strategy evaluate these solutions. The purpose is to demonstrate that both variants implemented without the arc-length procedure fail to follow the equilibrium curve throughout the fracturing process. Consequently, the estimation of the energy release is significantly overestimated.

Variant I is an implicit staggered strategy with sub-iterations of the mechanical and the damage stages. At the

load step k , sub-iterations j are repeated until reaching, between two consecutive sub-iterations, $j-1$ and j , an increment of phase field satisfying: $\|\varphi_k^j - \varphi_k^{j-1}\|_\infty \leq 0.01$. If this criterion is not satisfied in 100 sub-iterations, the scheme continues with the next load step. Variant II is an explicit staggered strategy with no sub-iterations, and $j = 1$. In all these cases, the loading process is defined by imposing fixed increments of vertical displacement $\Delta\delta$.

Figure 12 plots the total force f versus vertical displacement δ curves. The plot B corresponds to solutions obtained with variant I and an imposed displacement increment $\Delta\delta/b = 25 \times 10^{-6}$. Plots C and D correspond to solutions obtained with variant II and imposed displacement increments $\Delta\delta/b = 5 \times 10^{-6}$ and $\Delta\delta/b = 2.5 \times 10^{-6}$, respectively. These plots are compared with the solution A presented in Sect. 4.2 with the tolerance parameter $\Delta\hat{\varphi}^{opt} = 10^{-2}$.

After crossing the peak load, solution B with the maximum sub-iterations fixed at 100, is determined with only two points (1 and 2 in Fig. 12). This staggered scheme variant lacks parameters controlling the crack tip advance during this loading process stage. Therefore, the intermediate structural equilibrium points are unable to be reached. The numerical consequence of this effect is relevant when the energy balance condition is examined. The energy balance condition accuracy is assessed through the relative percent error of the energy release, e_W , defined in Eq. (51), and is compared for the solutions reported in this Section in the fourth column of Table 7. Additionally, Table 7 describes computational costs, in terms of the number of total iterations of the mechanical stage.

References

- Bourdin B, Francfort GA, Marigo JJ (2000) Numerical experiments in revisited brittle fracture. *J Mech Phys Solids* 48(4):797–826
- Farrell P, Maurini C (2017) Linear and nonlinear solvers for variational phase-field models of brittle fracture. *Int J Numer Methods Eng* 109(5):648–667
- Gerasimov T, De Lorenzis L (2016) A line search assisted monolithic approach for phase-field computing of brittle fracture. *Comput Methods Appl Mech Eng* 312:276–303
- Kristensen PK, Martínez-Pañeda E (2020) Phase field fracture modelling using quasi-newton methods and a new adaptive step scheme. *Theor Appl Fract Mech* 107:102446
- Wick T (2017) Modified newton methods for solving fully monolithic phase-field quasi-static brittle fracture propagation. *Comput Methods Appl Mech Eng* 325:577–611
- Heister T, Wheeler MF, Wick T (2015) A primal-dual active set method and predictor-corrector mesh adaptivity for computing fracture propagation using a phase-field approach. *Comput Methods Appl Mech Eng* 290:466–495
- Lampron O, Theriault D, Lévesque M (2021) An efficient and robust monolithic approach to phase-field quasi-static brittle fracture using a modified newton method. *Comput Methods Appl Mech Eng* 386:114091
- Kopaničáková A, Kothari H, Krause R (2023) Nonlinear field-split preconditioners for solving monolithic phase-field models of brittle fracture. *Comput Methods Appl Mech Eng* 403:115733
- Miehe C, Welschinger F, Hofacker M (2010) Thermodynamically consistent phase-field models of fracture: variational principles and multi-field Fe implementations. *Int J Numer Methods Eng* 83(10):1273–1311
- Bourdin B (2007) Numerical implementation of the variational formulation for quasi-static brittle fracture. *Interfaces Free bound* 9(3):411–430
- Burke S, Ortner Ch, Süli E (2010) An adaptive finite element approximation of a variational model of brittle fracture. *SIAM J Numer Anal* 48(3):980–1012
- Brun MK, Wick T, Berre I, Nordbotten JM, Radu FA (2020) An iterative staggered scheme for phase field brittle fracture propagation with stabilizing parameters. *Comput Methods Appl Mech Eng* 361:112752
- Duda FP, Barbosa JM, Guimarães LJ, Souza AC (2007) Modeling of coupled deformation-diffusion-damage in elastic solids. *Int J Model Simul Pet Ind* 1(1):85–93
- da Silva Jr MN, Duda FP, Fried E (2013) Sharp-crack limit of a phase-field model for brittle fracture. *J Mech Phys Solids* 61(11):2178–2195
- Duda FP, Ciarbonetti A, Sánchez PJ, Huespe AE (2015) A phase-field/gradient damage model for brittle fracture in elastic-plastic solids. *Int J Plast* 65:269–296
- Gutiérrez MA (2004) Energy release control for numerical simulations of failure in quasi-brittle solids. *Commun Numer Methods Eng* 20(1):19–29
- Verhoosel CV, Remmers JJC, Gutiérrez MA (2009) A dissipation-based arc-length method for robust simulation of brittle and ductile failure. *Int J Numer Methods Eng* 77(9):1290–1321
- Singh N, Verhoosel CV, De Borst R, Van Brummelen EH (2016) A fracture-controlled path-following technique for phase-field modeling of brittle fracture. *Finite Elem Anal Des* 113:14–29
- Bharali R, Goswami S, Anitescu C, Rabczuk T (2022) A robust monolithic solver for phase-field fracture integrated with fracture energy based arc-length method and under-relaxation. *Comput Methods Appl Mech Eng* 394:114927
- Börjesson E, Remmers JJC, Fagerström M (2022) A generalised path-following solver for robust analysis of material failure. *Comput Mech* 70(2):437–450
- Brust JJ, Marcia CG, Petra RF (2019) Large-scale quasi-Newton trust-region methods with low-dimensional linear equality constraints. *Comput Optim Appl* 74(3):669–701
- Martínez-Pañeda E, Busto SD, Betegón C (2017) Non-local plasticity effects on notch fracture mechanics. *Theor Appl Fract Mech* 92:276–287
- Wu JY (2018) Robust numerical implementation of non-standard phase-field damage models for failure in solids. *Comput Methods Appl Mech Eng* 340:767–797
- Tanné E, Li T, Bourdin B, Marigo JJ, Maurini C (2018) Crack nucleation in variational phase-field models of brittle fracture. *J Mech Phys Solids* 110:80–99
- Miehe C, Hofacker M, Welschinger F (2010) A phase field model for rate-independent crack propagation: robust algorithmic implementation based on operator splits. *Comput Methods Appl Mech Eng* 199(45–48):2765–2778
- Amor H, Marigo JJ, Maurini C (2009) Regularized formulation of the variational brittle fracture with unilateral contact: numerical experiments. *J Mech Phys Solids* 57(8):1209–1229
- Gerasimov T, De Lorenzis L (2019) On penalization in variational phase-field models of brittle fracture. *Comput Methods Appl Mech Eng* 354:990–1026
- Lu Y, Helfer T, Bary B, Fandeur O (2020) An efficient and robust staggered algorithm applied to the quasi-static description of brittle

- fracture by a phase-field approach. *Comput Methods Appl Mech Eng* 370:113218
29. May S, Vignollet J, de Borst R (2016) A new arc-length control method based on the rates of the internal and the dissipated energy. *Eng Comput* 33(1):100–115
 30. Crisfield MA (2000) *Non-linear finite element analysis of solids and structures*, vol 1. Wiley, Essential
 31. Pham K, Marigo JJ (2009) Construction and analysis of localized responses for gradient damage models in a 1d setting. *Vietnam J Mech* 31(3–4):233–246
 32. Ambati M, Gerasimov T, De Lorenzis L (2015) A review on phase-field models of brittle fracture and a new fast hybrid formulation. *Comput Mech* 55(2):383–405
 33. Kumar A, Bourdin B, Francfort GA, Lopez-Pamies O (2020) Revisiting nucleation in the phase-field approach to brittle fracture. *J Mech Phys Solids* 142:104027
 34. Li S, Thouless MD, Waas AM, Schroeder JA, Zavattieri PD (2005) Use of mode-I cohesive-zone models to describe the fracture of an adhesively-bonded polymer-matrix composite. *Compos Sci Technol* 65(2):281–293
 35. Zambrano J, Toro S, Sánchez PJ, Duda FP, Méndez CG, Huespe AE (2022) Interaction analysis between a propagating crack and an interface: phase field and cohesive surface models. *Int J Plast* 156:103341
 36. Wu JY, Huang Y, Nguyen VP (2020) On the BFGS monolithic algorithm for the unified phase field damage theory. *Comput Methods Appl Mech Eng* 360:112704

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.